MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR POWER PLANT DUCT SUPPORT

DESIGN CALCULATION OF FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-2002

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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for Design Calculation of Foundation for Power Plant Duct Support.

2. SCOPE

C01

The document is prepared for Design Calculation of Column Foundation for Power Plant Duct Support. Five types of Foundation namely F1 to F5 are designed for all the supports.
Load data for foundation is attached with this document. Refer Structural GA drawing MD-216-49PR-EG-ST-GAD-2406 & -2416 for location & details of duct supports.
Foundations are modelled & analysed in Staad-Pro. Support Reactions from Superstructure have been applied as loads to Foundations. As Seismic load is less then wind load
so foundations are designed for wind load. Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008 (as supported by STAAD-Pro).
Net Safe Bearing Capacility is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

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3 Design of Foundation for Duct Supporting Tower

3.1 Design of Foundation mkd. F1

C01 (Refer Load Data for Tower marked TW-1 & TW-1A)

	145			146			220			222		
LOAD ITEM	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton
1 DL	0.0	-0.9	0.0	0.0	-0.9	0.0	0.0	-0.9	0.0	0.0	-0.9	0.0
2 LL	0.9	-8.1	-3.0	0.9	12.9	0.0	0.0	-14.1	-3.0	0.0	6.5	0.0
3 WL1(+X)	0.6	1.0	0.0	0.6	1.0	0.0	0.0	-1.0	0.0	0.0	-1.0	0.0
4 WL2(-X)	-0.6	-1.0	0.0	-0.6	-1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0
5 WL3(-X)	0.0	3.3	1.2	0.0	-3.3	0.0	0.0	3.2	1.2	0.0	-3.2	0.0
6 WL4(-X)	0.0	-3.3	-1.2	0.0	3.3	0.0	0.0	-3.2	-1.2	0.0	3.2	0.0

C01

FOUNDATION SIZE (F1)	3.00m	X	3.00m
DEPTH OF THE FOUNDATION	2.50m		from FFL
THE WT OF SOIL +SELF WT	3x3x2.5x2 =		45 T

WORKING STRESS DESIGN	145			146			220			222		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+LL	0.88	-9.01	-3.03	0.92	12.02	0.01	0.00	-15.02	-3.00	0.00	5.55	0.02
DL+0.75*LL+0.75*WL1	1.08	-6.24	-2.27	1.11	9.51	0.01	0.01	-12.25	-2.25	0.01	3.20	0.02
DL+0.75*LL+0.75*WL2	0.24	-7.74	-2.27	0.28	8.05	0.01	-0.01	-10.75	-2.25	-0.02	4.66	0.01
DL+0.75*LL+0.75*WL3	0.67	-4.52	-1.36	0.69	6.32	0.02	0.00	-9.11	-1.35	0.00	1.54	0.02
DL+0.75*LL+0.75*WL4	0.65	-9.45	-3.18	0.70	11.24	0.00	0.00	-13.89	-3.15	0.00	6.32	0.01
0.6*DL+WL1	0.56	0.45	0.00	0.55	0.42	0.00	0.01	-1.56	0.00	0.02	-1.53	0.00
0.6*DL+WL2	-0.56	-1.56	0.00	-0.55	-1.53	0.00	-0.01	0.45	0.00	-0.02	0.42	0.00
0.6*DL+WL3	0.01	2.73	1.21	-0.01	-3.84	0.01	0.00	2.63	1.20	0.00	-3.74	0.01
0.6*DL+WL4	-0.01	-3.84	-1.21	0.01	2.73	-0.01	0.00	-3.74	-1.20	0.00	2.63	-0.01

SLIDING AND OVERTURNING CHECK

	Load at bottom of foundation					OVERTURNING CHECK		SLIDING CHECK, m = 0.4	
LOAD CASES	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	51.47	1.8	49.2	-6.00	14.76	5.23	1.57	11.44	3.43
DL+0.75*LL+0.75*WL1	50.78	2.2	36.9	-4.50	15.86	4.80	2.06	9.21	4.51
DL+0.75*LL+0.75*WL2	50.78	0.5	36.9	-4.50	6.28	12.13	2.06	41.11	4.51
DL+0.75*LL+0.75*WL3	50.78	1.4	24.1	-2.67	11.07	6.88	3.15	15.04	7.60
DL+0.75*LL+0.75*WL4	50.78	1.4	49.7	-6.33	11.07	6.88	1.53	15.04	3.21
0.6*DL+WL1	47.22	1.1	0.0	0.00	6.39	11.09	32794.00	16.55	26235.20
0.6*DL+WL2	47.22	-1.1	0.0	0.00	6.39	11.09	32794.00	16.55	26235.20
0.6*DL+WL3	47.22	0.0	17.0	2.44	0.00	N.A	4.16	N.A	7.75
0.6*DL+WL4	47.22	0.0	17.0	-2.44	0.00	N.A	4.16	N.A	7.75

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 23T/m^2 IN LC 1. HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

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Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 109)

REINFORCEMENT CALCULATION OF FOUNDATION:

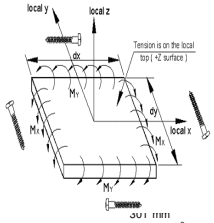
Foundation Thickness =	400	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	3.886	T-M	Corresponding Mxy	-0.83	T-M	*(Plate No. - 139 , L/C No. - 101)*
Min. Mx	-1.34	T-M	Corresponding Mxy	-0.84	T-M	*(Plate No. - 82 , L/C No. - 101)*
Max. My	3.668	T-M	Corresponding Mxy	-1.33	T-M	*(Plate No. - 138 , L/C No. - 101)*
Min. My	-4.754	T-M	Corresponding Mxy	-2.22	T-M	*(Plate No. - 385 , L/C No. - 101)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	3.886 + 0.833 =	4.719 T-M
My1 =	My + abs(Mxy) =	3.668 + 1.328 =	4.996 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	3.886 + 0.189 =	4.075 T-M
My2 =	My+abs(Mxy^2/Mx) =	3.668 + 0.454 =	4.122 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 4.719 T-m
Myd = 4.996 T-m

ALONG X - DIRECTION :

Mu = 4.719 T-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 4.719X10^7/(1000X301^2) = 0.521 N/mm^2 < 6.9 N/mm^2
Pt = 0.0014 Reqd. Ast = 0.0014X1000X301/100 = 4.21 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 4.996 T-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 4.996X10^7/(1000X317^2) = 0.497 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X317/100 = 4.12 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-1.34 - 0.835 =	-2.175 T-M
My1 =	My - abs(Mxy) =	-4.754 - 2.216 =	-6.97 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-1.34 - 0.147 =	-1.487 T-M
My2 =	My-abs(Mxy^2/Mx) =	-4.754 - 3.665 =	-1.089 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -2.175 T-M
Myd = -6.97 T-M

ALONG X - DIRECTION :

Mu = 2.175 T-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 2.175X10^7/(1000X301^2) = 0.24 N/mm^2 < 6.9 N/mm^2
Pt = 0.0006 Reqd. Ast = 0.0006X1000X301/100 = 1.81 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 6.97 T-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 6.97X10^7/(1000X317^2) = 0.694 N/mm^2 < 6.9 N/mm^2
Pt = 0.0019 Reqd. Ast = 0.0019X1000X317/100 = 6.02 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

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SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.239 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 392 , L/C No. - 101)*
Vu = 7.194 T

Provided Pt = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X7.194X10^4X301/2.175/10^7)X1000X301/10^4
= 30.173 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^4
= 30.18 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^4
= 51.641 T
HENCE, Vc = 30.173 T
Φt*Vc = 0.75X30.173 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 22.63 T **> 7.194T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.247 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 619 , L/C No. - 101)*
Vu = 7.83 T

Pt provided = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X7.83X10^4X317/6.97/10^7)X1000X317/10^4
= 30.62 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^4
= 31.731 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^4
= 54.387 T
HENCE, Vc = 30.62 T
Φt*Vc = 0.75X30.62 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 22.965 T **> 7.83T , Hence O.K.**

DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F1

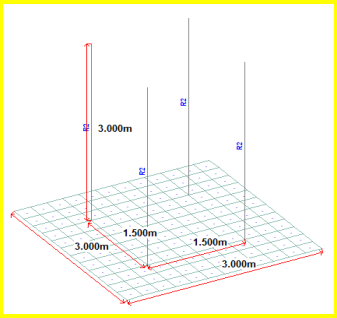
Size of Column = 550 X 550 mm^2

Reqired area of Reinforcement = 3025 mm^2

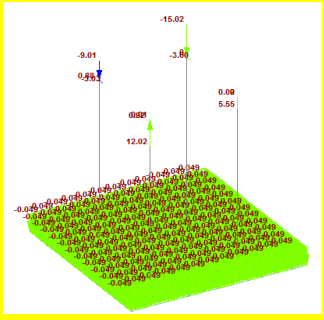
Reinforcement Provided = 4 no. 20 dia bars + 4 no. 25 dia bars

Area of Reinforcement Provided = 3220.1 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



C01

Surcharge Load on Foundation = 1 T/m^2
Soil Weight over the foundation = 2.1x 1.8 T/m2 = 3.78 T/m^2
SOIL+SURCHARGE LOAD = 4.78 T/m^2

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C01

3.2 Design of Foundation mkd. F2 (Refer Load Data for Tower marked TW-2)

	145			146			220			222		
LOAD ITEM	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton
1 DL	0.0	-0.9	0.0	0.0	-0.9	0.0	0.0	-0.7	0.0	0.0	-0.7	0.0
2 LL	0.7	7.6	1.7	-1.4	-10.9	0.0	0.0	10.6	3.9	0.0	-9.9	0.0
3 WL1(+X)	0.6	0.9	0.0	0.6	0.9	0.0	0.0	-0.9	0.0	0.0	-0.9	0.0
4 WL2(-X)	-0.6	-0.9	0.0	-0.6	-0.9	0.0	0.0	0.9	0.0	0.0	0.9	0.0
5 WL3(-X)	0.0	3.3	1.2	0.0	-3.3	0.0	0.0	3.2	1.2	0.0	-3.2	0.0
6 WL4(-X)	0.0	-3.3	-1.2	0.0	3.3	0.0	0.0	-3.2	-1.2	0.0	3.2	0.0

FOUNDATION SIZE (F2) 2.50m X 3.50m
DEPTH OF THE FOUNDATION 2.50m from FFL
THE WT OF SOIL +SELF WT 2.5x3.5x2.5x2 = 43.75 T

WORKING STRESS DESIGN	145			146			220			222		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+LL	0.74	6.65	1.73	-1.41	-11.83	-0.01	0.00	9.91	3.92	0.00	-10.63	-0.01
DL+0.75*LL+0.75*WL1	0.97	5.42	1.30	-0.65	-8.46	-0.01	0.01	6.59	2.94	0.01	-8.80	-0.01
DL+0.75*LL+0.75*WL2	0.14	4.10	1.30	-1.48	-9.75	-0.01	-0.01	7.92	2.94	-0.01	-7.50	-0.01
DL+0.75*LL+0.75*WL3	0.56	7.22	2.21	-1.07	-11.57	0.00	0.00	9.64	3.84	0.00	-10.54	0.00
DL+0.75*LL+0.75*WL4	0.55	2.30	0.39	-1.05	-6.64	-0.02	0.00	4.86	2.04	0.00	-5.76	-0.02
0.6*DL+WL1	0.56	0.34	0.00	0.55	0.32	0.00	0.01	-1.32	0.00	0.02	-1.29	0.00
0.6*DL+WL2	-0.56	-1.43	0.00	-0.55	-1.40	0.00	-0.01	0.45	0.00	-0.02	0.43	0.00
0.6*DL+WL3	0.01	2.74	1.21	-0.01	-3.83	0.01	0.00	2.75	1.20	0.00	-3.62	0.01
0.6*DL+WL4	-0.01	-3.83	-1.21	0.01	2.74	-0.01	0.00	-3.62	-1.20	0.00	2.75	-0.01

SLIDING AND OVERTURNING CHECK

	Load at bottom of foundation					OVERTURNING CHECK		SLIDING CHECK, μ = 0.4	
LOAD CASES	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	49.65	-0.7	46.1	5.63	5.82	10.67	1.88	29.55	3.53
DL+0.75*LL+0.75*WL1	48.99	0.4	34.6	4.22	1.76	34.89	2.48	55.83	4.64
DL+0.75*LL+0.75*WL2	48.99	-1.4	34.6	4.22	9.23	6.64	2.48	14.42	4.64
DL+0.75*LL+0.75*WL3	48.99	-0.5	47.4	6.05	4.44	13.80	1.81	38.88	3.24
DL+0.75*LL+0.75*WL4	48.99	-0.5	21.8	2.39	4.44	13.80	3.92	38.88	8.19
0.6*DL+WL1	45.71	1.1	0.0	0.00	6.20	9.21	37029.94	16.04	25391.96
0.6*DL+WL2	45.71	-1.1	0.0	0.00	6.57	8.70	37029.94	16.04	25391.96
0.6*DL+WL3	45.71	0.0	17.0	2.44	0.18	311.18	4.70	N.A	7.51
0.6*DL+WL4	45.71	0.0	17.0	-2.44	0.18	311.18	4.70	N.A	7.50

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 17T/m^2 IN LC 1. HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED (01)	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED (01)	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 109)

REINFORCEMENT CALCULATION OF FOUNDATION:

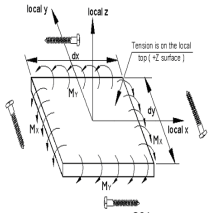
Foundation Thickness =	400	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	2.399	T-M	Corresponding Mxy	-0.02	T-M	*(Plate No. - 75 , L/C No. - 102)*
Min. Mx	-4.424	T-M	Corresponding Mxy	-0.17	T-M	*(Plate No. - 27 , L/C No. - 101)*
Max. My	9.265	T-M	Corresponding Mxy	-1.64	T-M	*(Plate No. - 385 , L/C No. - 101)*
Min. My	-6.133	T-M	Corresponding Mxy	-1.32	T-M	*(Plate No. - 388 , L/C No. - 101)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	2.399 + 0.024 =	2.423	T-M
My1 =	My + abs(Mxy) =	9.265 + 1.636 =	10.901	T-M
Mx2 =	Mx+abs(Mxy^2/My) =	2.399 + 0 =	2.399	T-M
My2 =	My+abs(Mxy^2/Mx) =	9.265 + 1.116 =	10.381	T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 2.423 T-m
Myd = 10.901 T-m

ALONG X - DIRECTION :

Mu = 2.423 T-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 2.423X10^7/(1000X301^2) = 0.267 N/mm^2 < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast = 0.0007X1000X301/100 = 2.11 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 10.901 T-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 10.901X10^7/(1000X317^2) = 1.085 N/mm^2 < 6.9 N/mm^2
Pt = 0.0029 Reqd. Ast = 0.0029X1000X317/100 = 9.19 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-4.424 - 0.172 =	-4.596	T-M
My1 =	My - abs(Mxy) =	-6.133 - 1.323 =	-7.456	T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-4.424 - 0.005 =	-4.429	T-M
My2 =	My-abs(Mxy^2/Mx) =	-6.133 - 0.396 =	-5.737	T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -4.596 T-M
Myd = -7.456 T-M

ALONG X - DIRECTION :

Mu = 4.596 T-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 4.596X10^7/(1000X301^2) = 0.507 N/mm^2 < 6.9 N/mm^2
Pt = 0.0014 Reqd. Ast = 0.0014X1000X301/100 = 4.21 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 7.456 T-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 7.456X10^7/(1000X317^2) = 0.742 N/mm^2 < 6.9 N/mm^2
Pt = 0.0020 Reqd. Ast = 0.002X1000X317/100 = 6.34 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023	Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 001	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.176 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 467 , L/C No. - 101)*
Vu = 5.298 T

Provided Pt = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X5.298X10^4X301/4.596/10^7)X1000X301/10^4
= 29.078 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^4
= 30.18 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^4
= 51.641 T
HENCE, Vc = 29.078 T
Φt*Vc = 0.75X29.078 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 21.809 T **> 5.298T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.389 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 94 , L/C No. - 101)*
Vu = 12.331 T

Pt provided = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X12.331X10^4X317/7.456/10^7)X1000X317/10^4
= 30.91 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^4
= 31.731 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^4
= 54.387 T
HENCE, Vc = 30.91 T
Φt*Vc = 0.75X30.91 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 23.183 T **> 12.331T , Hence O.K.**

DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F2

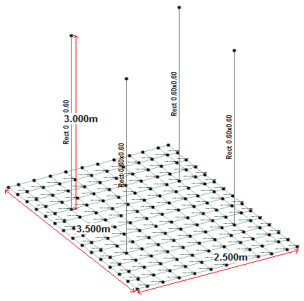
Size of Column = 550 X 550 mm^2

Reqired area of Reinforcement = 3025 mm^2

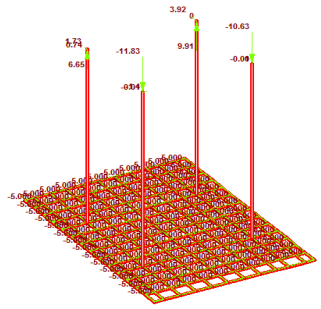
Reinforcement Provided = 4 no. 20 dia bars + 4 no. 25 dia bars

Area of Reinforcement Provided = 3220.1 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2
Soil Weight over the foundation = 2.1x 1.8 T/m2 = 3.8 T/m^2
SOIL+SURCHARGE LOAD = 4.78 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

C01

3.3 Design of Foundation mkd. F3 (Refer Load Data for 4 L Tower)

	145			146			220			222		
LOAD ITEM	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton
1 DL	-0.1	-2.5	0.0	0.0	-1.7	0.0	0.0	-3.0	0.0	0.0	-2.1	0.0
2 LL (including Pipe Load)	3.2	21.2	0.0	-0.2	8.2	0.0	2.4	-18.0	-6.9	0.1	-35.1	-7.3
3 WL1(+X)	1.1	2.0	0.0	0.0	-2.0	0.0	1.2	2.0	0.0	0.0	-2.0	0.0
4 WL2(-X)	-1.1	-2.0	0.0	0.0	2.0	0.0	-1.2	-2.0	0.0	0.0	2.0	0.0
5 WL3(-X)	-0.1	-2.8	0.0	0.0	-3.0	0.0	0.1	2.8	1.2	0.0	3.0	1.2
6 WL4(-X)	0.1	2.8	0.0	0.0	3.0	0.0	-0.1	-2.8	-1.2	0.0	-3.0	-1.2

FOUNDATION SIZE (F1) 3.35m X 3.35m
DEPTH OF THE FOUNDATION 2.50m from FFL
THE WT OF SOIL +SELF WT 3.35x3.35x2.5x2 = 56.1125 T

WORKING STRESS DESIGN	145			146			220			222		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+LL	3.11	18.74	0.00	-0.18	6.49	0.01	2.40	-21.00	-6.87	0.08	-37.23	-7.32
DL+0.75*LL+0.75*WL1	3.17	14.93	0.00	-0.12	2.96	0.01	2.69	-14.98	-5.15	0.08	-29.97	-5.50
DL+0.75*LL+0.75*WL2	1.46	11.96	0.00	-0.14	5.92	0.01	0.92	-18.01	-5.15	0.05	-26.94	-5.49
DL+0.75*LL+0.75*WL3	2.28	11.33	0.02	-0.12	2.16	0.03	1.85	-14.38	-4.22	0.05	-26.18	-4.56
DL+0.75*LL+0.75*WL4	2.35	15.56	-0.02	-0.14	6.73	-0.01	1.76	-18.62	-6.08	0.08	-30.74	-6.43
0.6*DL+WL1	1.09	0.50	0.00	0.03	-3.00	0.00	1.19	0.22	0.02	0.03	-3.30	-0.02
0.6*DL+WL2	-1.20	-3.45	0.00	0.00	0.95	0.00	-1.18	-3.82	0.01	0.00	0.74	0.00
0.6*DL+WL3	-0.10	-4.30	0.03	0.03	-4.07	0.03	0.06	1.03	1.25	0.00	1.76	1.23
0.6*DL+WL4	0.00	1.35	-0.02	-0.01	2.02	-0.02	-0.06	-4.63	-1.22	0.03	-4.32	-1.25

SLIDING AND OVERTURNING CHECK

	Load at bottom of foundation					OVERTURNING CHECK		SLIDING CHECK, m = 0.4	
LOAD CASES	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	89.11	5.4	66.8	-14.18	87.19	1.71	2.24	6.58	2.51
DL+0.75*LL+0.75*WL1	83.18	5.8	54.8	-10.64	70.88	1.97	2.54	5.72	3.13
DL+0.75*LL+0.75*WL2	83.18	2.3	44.6	-10.63	60.26	2.31	3.12	14.58	3.13
DL+0.75*LL+0.75*WL3	83.18	4.1	44.0	-8.73	58.10	2.40	3.16	8.20	3.81
DL+0.75*LL+0.75*WL4	83.18	4.0	55.4	-12.53	73.04	1.91	2.51	8.22	2.66
0.6*DL+WL1	61.69	2.3	6.0	0.00	7.50	13.78	17.28	10.57	7908.93
0.6*DL+WL2	61.69	-2.4	7.7	0.01	7.64	13.52	13.50	10.35	1939.93
0.6*DL+WL3	61.69	0.0	8.4	2.54	9.54	10.83	12.26	1370.88	9.72
0.6*DL+WL4	61.69	0.0	8.4	-2.52	10.57	9.78	12.30	761.60	9.78

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 28T/m^2 IN LC 1. HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED (001)	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED (001)	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 109)

REINFORCEMENT CALCULATION OF FOUNDATION:

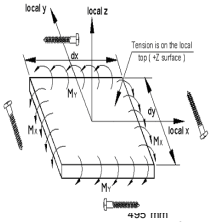
Foundation Thickness =	600	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	7.522	T-M	Corresponding Mxy	6.56	T-M	*(Plate No. - 623 , L/C No. - 101)*
Min. Mx	-5.927	T-M	Corresponding Mxy	2.34	T-M	*(Plate No. - 130 , L/C No. - 101)*
Max. My	20.078	T-M	Corresponding Mxy	6.56	T-M	*(Plate No. - 623 , L/C No. - 101)*
Min. My	-13.348	T-M	Corresponding Mxy	6.18	T-M	*(Plate No. - 345 , L/C No. - 101)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	7.522 + 6.56 =	14.082 T-M
My1 =	My + abs(Mxy) =	20.078 + 6.56 =	26.638 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	7.522 + 2.143 =	9.665 T-M
My2 =	My+abs(Mxy^2/Mx) =	20.078 + 5.721 =	25.799 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 14.082 T-m
Myd = 26.638 T-m

ALONG X - DIRECTION :

Mu = 14.082 T-m
Φ*Kn = Mu/bd^2 deff. = 495 mm
Mu/bd^2 = 14.082X10^7/(1000X495^2) = 0.575 N/mm^2 < 6.9 N/mm^2
Pt = 0.0015 Reqd. Ast = 0.0015X1000X495/100 = 7.43 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 26.638 T-m
Φ*Kn = Mu/bd^2 deff. = 515 mm
Mu/bd^2 = 26.638X10^7/(1000X515^2) = 1.004 N/mm^2 < 6.9 N/mm^2
Pt = 0.0027 Reqd. Ast = 0.0027X1000X515/100 = 13.91 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-5.927 - 2.335 =	-8.262 T-M
My1 =	My - abs(Mxy) =	-13.348 - 6.18 =	-19.528 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-5.927 - 0.408 =	-6.335 T-M
My2 =	My-abs(Mxy^2/Mx) =	-13.348 - 6.444 =	-6.904 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -8.262 T-M
Myd = -19.528 T-M

ALONG X - DIRECTION :

Mu = 8.262 T-m
Φ*Kn = Mu/bd^2 deff. = 495 mm
Mu/bd^2 = 8.262X10^7/(1000X495^2) = 0.337 N/mm^2 < 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast = 0.0009X1000X495/100 = 4.46 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 19.528 T-m
Φ*Kn = Mu/bd^2 deff. = 515 mm
Mu/bd^2 = 19.528X10^7/(1000X515^2) = 0.736 N/mm^2 < 6.9 N/mm^2
Pt = 0.0020 Reqd. Ast = 0.002X1000X515/100 = 10.3 cm^2
Provided **20** mm @ **200** C/C (Provide Ast = 15.71 cm2) at Horizontal

Total Provide Ast = 15.71 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023	Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 001	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.555 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 658 , L/C No. - 101)*
Vu = 27.473 T

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X27.473X10^4X495/8.262/10^7)X1000X495/10^4
= 51.288 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X495/10^4
= 49.548 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X495/10^4
= 84.925 T
HENCE, Vc = 49.548 T
Φt*Vc = 0.75X49.548 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 37.161 T **> 27.473T , Hence O.K.**

ALONG Y - DIRECTION :

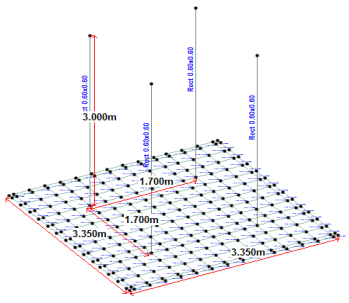
SQy = -0.443 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 674 , L/C No. - 101)*
Vu = 22.815 T

Pt provided = 0.0031
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0031X22.815X10^4X515/19.528/10^7)X1000X515/10^4
= 50.382 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0031)X1000X515/10^4
= 51.463 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X515/10^4
= 88.357 T
HENCE, Vc = 50.382 T
Φt*Vc = 0.75X50.382 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 37.787 T **> 22.815T , Hence O.K.**

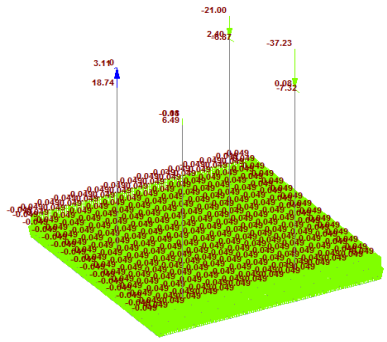
DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F3

Size of Column = 600 X 600 mm^2
Reqired area of Reinforcement = 4932 mm^2
Reinforcement Provided = 16 no. 20 dia bars
Area of Reinforcement Provided = 5026.5 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2
Soil Weight over the foundation = 2.1x 1.8 T/m2 = 3.78 T/m^2
SOIL+SURCHARGE LOAD = 4.78 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
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Date	04-06-2020	07-08-2020	26-10-2023
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Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

C01

3.4 Design of Foundation mkd. F4

(Refer Load Data For 2 Legged Trestle on Grid 1 & 14)

	C-1			C-2		
LOAD ITEM	Fx (T)	Fy (T)	Fz (T)	Fx (T)	Fy (T)	Fz (T)
1 DL	-0.01	-2.33	0.00	0.02	-0.42	0.00
2 LL	0.60	-1.17	0.01	0.00	-2.90	0.00
21 WL1(+X)	1.10	1.77	0.00	0.02	-1.77	0.00
22 WL2(-X)	-1.10	-1.77	0.00	-0.02	1.77	0.06
23 WL3(+Z)	0.04	0.07	0.06	0.00	-0.07	-0.06
24 WL4(-Z)	-0.04	-0.07	-0.06	0.00	0.07	0.00

FOUNDATION SIZE 1.50m X 2.50m
DEPTH OF THE FOUNDATION 2.50m (below FGL)
THE WT OF SOIL +SELF WT 1.5X2.5X2.5XUNIT WT = 18.75 T

WORKING STRESS DESIGN

	C-1			C-2			Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	0.59	-3.50	0.01	0.02	-3.32	0.00	25.56	0.61	0.02	0.01	1.98
DL+0.75*LL+0.75*WL1	1.27	-1.88	0.00	0.03	-3.92	0.00	24.55	1.30	0.01	0.00	5.62
DL+0.75*LL+0.75*WL2	-0.38	-4.54	0.00	0.00	-1.26	0.04	24.55	-0.38	0.14	0.05	3.92
DL+0.75*LL+0.75*WL3	0.47	-3.16	0.05	0.02	-2.64	-0.04	24.55	0.49	0.02	0.01	1.90
DL+0.75*LL+0.75*WL4	0.41	-3.26	-0.04	0.02	-2.54	0.00	24.55	0.43	0.11	-0.04	1.89
0.6*DL+WL1	1.09	0.38	0.00	0.03	-2.02	0.00	20.39	1.12	0.00	0.00	5.40
0.6*DL+WL2	-1.10	-3.17	0.00	-0.01	1.52	0.06	20.39	-1.11	0.17	0.06	7.32
0.6*DL+WL3	0.04	-1.33	0.06	0.01	-0.32	-0.06	20.39	0.05	0.00	0.00	1.00
0.6*DL+WL4	-0.04	-1.46	-0.06	0.01	-0.18	0.00	20.39	-0.03	0.16	-0.05	1.19

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	25.56	0.61	1.98	0.01	0.02	16.13	-	16.81	1704.29
DL+0.75*LL+0.75*WL1	24.55	1.30	5.62	0.00	0.01	5.46	-	7.58	2181.87
DL+0.75*LL+0.75*WL2	24.55	-0.38	3.92	0.05	0.14	7.83	133.69	25.95	213.91
DL+0.75*LL+0.75*WL3	24.55	0.49	1.90	0.01	0.02	16.11	-	20.10	1818.22
DL+0.75*LL+0.75*WL4	24.55	0.43	1.89	-0.04	0.11	16.20	170.46	22.89	272.73
0.6*DL+WL1	20.39	1.12	5.40	0.00	0.00	4.72	-	7.27	-
0.6*DL+WL2	20.39	-1.11	7.32	0.06	0.17	3.48	92.37	7.35	147.79
0.6*DL+WL3	20.39	0.05	1.00	0.00	0.00	25.54	-	-	6798.16
0.6*DL+WL4	20.39	-0.03	1.19	-0.05	0.16	21.46	94.42	-	151.07

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 13T/m^2 IN LC 7 . HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023	Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 001	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 109) (REFER STAAD MODEL)

REINFORCEMENT CALCULATION OF FOUNDATION:

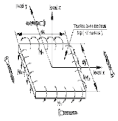
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	1.851	T-M	Corresponding Mxy	0.51	T-M	*(Plate No. - 137 , L/C No. - 102)*
Min. Mx	-1.457	T-M	Corresponding Mxy	0.86	T-M	*(Plate No. - 130 , L/C No. - 102)*
Max. My	0.955	T-M	Corresponding Mxy	0.90	T-M	*(Plate No. - 75 , L/C No. - 102)*
Min. My	-1.13	T-M	Corresponding Mxy	0.71	T-M	*(Plate No. - 42 , L/C No. - 102)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	1.851 + 0.507 =	2.358 T-M
My1 =	My + abs(Mxy) =	0.955 + 0.895 =	1.85 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	1.851 + 0.269 =	2.12 T-M
My2 =	My+abs(Mxy^2/Mx) =	0.955 + 0.433 =	1.388 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 2.358 T-m
Myd = 1.85 T-m

ALONG X - DIRECTION :

Mu = 2.358 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 2.358X10^7/(1000X207^2) = 0.55 N/mm^2 < 6.9 N/mm^2
Pt = 0.0015 Reqd. Ast = 0.0015X1000X207/100 = 3.11 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.85 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 1.85X10^7/(1000X219^2) = 0.386 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X219/100 = 2.19 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-1.457 - 0.858 =	-2.315 T-M
My1 =	My - abs(Mxy) =	-1.13 - 0.708 =	-1.838 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-1.457 - 0.651 =	-2.108 T-M
My2 =	My-abs(Mxy^2/Mx) =	-1.13 - 0.344 =	-0.786 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -2.315 T-M
Myd = -1.838 T-M

ALONG X - DIRECTION :

Mu = 2.315 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 2.315X10^7/(1000X207^2) = 0.54 N/mm^2 < 6.9 N/mm^2
Pt = 0.0014 Reqd. Ast = 0.0014X1000X207/100 = 2.9 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.838 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 1.838X10^7/(1000X219^2) = 0.383 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X219/100 = 2.19 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 001	AS BUILT
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Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
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SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.129 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 74 , L/C No. - 102)*
Vu = 2.67 T

Provided Pt = 0.0027
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0027X2.67X10^4X207/2.315/10^7)X1000X207/10^4
= 19.821 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0027)X1000X207/10^4
= 20.544 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X207/10^4
= 35.514 T
HENCE, Vc = 19.821 T
Φt*Vc = 0.75X19.821 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 14.866 T **> 2.67T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = 0.243 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 66 , L/C No. - 102)*
Vu = 5.322 T

Pt provided = 0.0026
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0026X5.322X10^4X219/1.838/10^7)X1000X219/10^4
= 21.344 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4
= 21.698 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X219/10^4
= 37.573 T
HENCE, Vc = 21.344 T
Φt*Vc = 0.75X21.344 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 16.008 T **> 5.322T , Hence O.K.**

PEDESTAL REINFORCEMENT

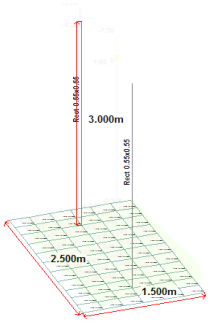
Size of Column = 550 X 550 mm^2

Reqired area of Reinforcement = 3025 mm^2

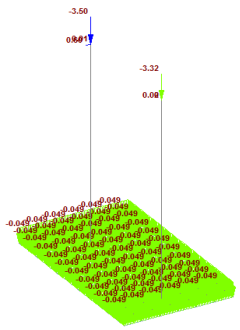
Reinforcement Provided = 4 no. 20 dia bars + 4 no. 25 dia bars

Area of Reinforcement Provided = 3220.1 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2
Soil Weight over the foundation = 2.2x 1.8 T/m2 = 3.96 T/m^2
SOIL+SURCHARGE LOAD = 4.96 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
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Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

C01

3.5 Design of Foundation mkd. F5

(Refer Load Data For 2 Legged Trestle on section G-G)

	C-1			C-2		
LOAD ITEM	Fx (T)	Fy (T)	Fz (T)	Fx (T)	Fy (T)	Fz (T)
1 DL	0.00	-2.00	0.00	0.00	-0.54	0.00
2 LL	-0.15	-3.70	0.00	1.60	-7.00	0.00
21 WL1(+X)	0.02	1.04	0.00	0.41	-1.04	0.00
22 WL2(-X)	-0.02	-1.04	0.00	-0.41	1.04	0.00
23 WL3(+Z)	0.00	0.00	0.00	0.00	0.00	0.00
24 WL4(-Z)	0.00	0.00	0.00	0.00	0.00	0.00

FOUNDATION SIZE 1.20m X 1.80m
DEPTH OF THE FOUNDATION 2.50m (below FGL)
THE WT OF SOIL +SELF WT 1.2X1.8X2.5XUNIT WT = 10.8 T

Eccentricity of the column with respect to foundation base = 200 mm

WORKING STRESS DESIGN

	C-1			C-2			Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	-0.15	-5.70	0.00	1.60	-7.54	0.00	24.04	1.45	2.65	0.00	5.21
DL+0.75*LL+0.75*WL1	-0.10	-3.99	0.00	1.51	-6.57	0.00	21.37	1.41	2.12	0.00	5.42
DL+0.75*LL+0.75*WL2	-0.13	-5.56	0.00	0.89	-5.01	0.00	21.37	0.77	2.12	0.00	2.56
DL+0.75*LL+0.75*WL3	-0.11	-4.78	0.00	1.20	-5.79	0.00	21.37	1.09	2.12	0.00	3.74
DL+0.75*LL+0.75*WL4	-0.11	-4.78	0.00	1.20	-5.79	0.00	21.37	1.09	2.12	0.00	3.74
0.6*DL+WL1	0.02	-0.16	0.00	0.41	-1.37	0.00	12.33	0.43	0.31	0.00	1.84
0.6*DL+WL2	-0.02	-2.24	0.00	-0.41	0.72	0.00	12.33	-0.43	0.31	0.00	2.66
0.6*DL+WL3	0.00	-1.20	0.00	0.00	-0.33	0.00	12.33	0.00	0.31	0.00	0.41
0.6*DL+WL4	0.00	-1.20	0.00	0.00	-0.33	0.00	12.33	0.00	0.31	0.00	0.41

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	24.04	1.45	5.21	0.00	2.65	4.16	5.44	6.63	7397.66
DL+0.75*LL+0.75*WL1	21.37	1.41	5.42	0.00	2.12	3.55	6.06	6.07	8766.11
DL+0.75*LL+0.75*WL2	21.37	0.77	2.56	0.00	2.12	7.53	6.06	11.14	8766.11
DL+0.75*LL+0.75*WL3	21.37	1.09	3.74	0.00	2.12	5.15	6.06	7.86	8766.11
DL+0.75*LL+0.75*WL4	21.37	1.09	3.74	0.00	2.12	5.15	6.06	7.86	8766.11
0.6*DL+WL1	12.33	0.43	1.84	0.00	0.31	6.01	24.24	11.54	-
0.6*DL+WL2	12.33	-0.43	2.66	0.00	0.31	4.17	24.24	11.54	-
0.6*DL+WL3	12.33	0.00	0.41	0.00	0.31	27.28	24.24	-	-
0.6*DL+WL4	12.33	0.00	0.41	0.00	0.31	27.28	24.24	-	-

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 18T/m^2 IN LC 2 . HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 001	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

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Date	04-06-2020	07-08-2020	26-10-2023
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Foundation has been designed with the Design strength Load Combinations of Staad File (LC 100 - LC 109) (REFER STAAD MODEL)

REINFORCEMENT CALCULATION OF FOUNDATION:

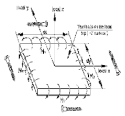
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	3.691	T-M	Corresponding Mxy	-0.64	T-M	*(Plate No. - 146 , L/C No. - 101)*
Min. Mx	-0.515	T-M	Corresponding Mxy	0.02	T-M	*(Plate No. - 145 , L/C No. - 107)*
Max. My	1.111	T-M	Corresponding Mxy	-0.74	T-M	*(Plate No. - 126 , L/C No. - 101)*
Min. My	-0.85	T-M	Corresponding Mxy	-0.90	T-M	*(Plate No. - 86 , L/C No. - 101)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	3.691 + 0.637 =	4.328 T-M
My1 =	My + abs(Mxy) =	1.111 + 0.74 =	1.851 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	3.691 + 0.365 =	4.056 T-M
My2 =	My+abs(Mxy^2/Mx) =	1.111 + 0.148 =	1.259 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 4.328 T-m
Myd = 1.851 T-m

ALONG X - DIRECTION :

Mu = 4.328 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 4.328X10^7/(1000X207^2) = 1.01 N/mm^2 < 6.9 N/mm^2
Pt = 0.0027 Reqd. Ast = 0.0027X1000X207/100 = 5.59 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Horizontal

Total Provide Ast = 7.54 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.851 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 1.851X10^7/(1000X219^2) = 0.386 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X219/100 = 2.19 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Horizontal

Total Provide Ast = 7.54 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-0.515 - 0.015 =	-0.53 T-M
My1 =	My - abs(Mxy) =	-0.85 - 0.9 =	-1.75 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-0.515 - 0 =	-0.515 T-M
My2 =	My-abs(Mxy^2/Mx) =	-0.85 - 1.573 =	0.723 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -0.53 T-M
Myd = -1.75 T-M

ALONG X - DIRECTION :

Mu = 0.53 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 0.53X10^7/(1000X207^2) = 0.124 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003 Reqd. Ast = 0.0003X1000X207/100 = 0.62 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Horizontal

Total Provide Ast = 7.54 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.75 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 1.75X10^7/(1000X219^2) = 0.365 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X219/100 = 2.19 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Horizontal

Total Provide Ast = 7.54 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	04-06-2020	07-08-2020	26-10-2023				Date	04-06-2020	07-08-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 101	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED 101	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.133 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 26 , L/C No. - 101)*
Vu = 2.753 T

Provided Pt = 0.0036
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0036X2.753X10^4X207/0.53/10^7)X1000X207/10^4
= 20.956 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0036)X1000X207/10^4
= 20.861 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X207/10^4
= 35.514 T
HENCE, Vc = 20.861 T
Φt*Vc = 0.75X20.861 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.646 T **> 2.753T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.161 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 34 , L/C No. - 101)*
Vu = 3.526 T

Pt provided = 0.0034
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0034X3.526X10^4X219/1.75/10^7)X1000X219/10^4
= 21.288 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0034)X1000X219/10^4
= 21.996 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X219/10^4
= 37.573 T
HENCE, Vc = 21.288 T
Φt*Vc = 0.75X21.288 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.966 T **> 3.526T , Hence O.K.**

PEDESTAL REINFORCEMENT

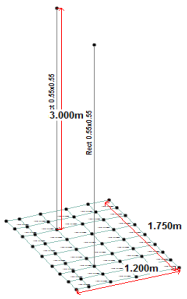
Size of Column = 550 X 550 mm^2

Reqired area of Reinforcement = 3025 mm^2

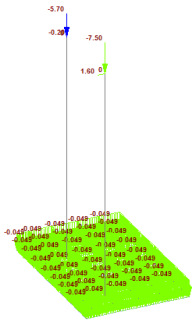
Reinforcement Provided = 4 no. 20 dia bars + 4 no. 25 dia bars

Area of Reinforcement Provided = 3220.1 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2
Soil Weight over the foundation = 2.2x 1.8 T/m2 = 3.96 T/m^2
SOIL+SURCHARGE LOAD = 4.96 T/m^2

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Date	04-06-2020	07-08-2020	26-10-2023			
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Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI			

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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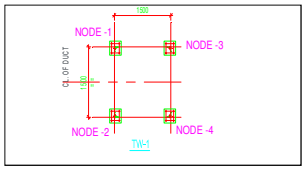
Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
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Prepared By	SHD	SHD	SHD
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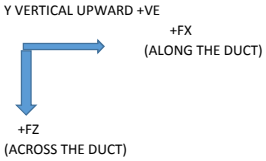
ATTACHMENT 1 Foundation Load Data for Duct Suppports

LOAD DATA : POWER PLANT DUCT SUPPORTING TOWER

TW-1

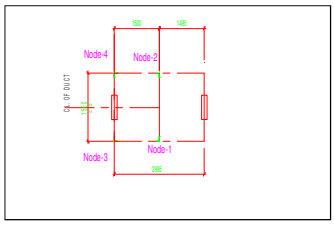
Node	L/C	Horizontal Fx Mton	Vertical Fy Mton	Horizontal Fz Mton
1	1 DL	0.0	-0.9	0.0
	2 LL	0.9	-8.1	-3.0
	3 WL+X	0.6	1.0	0.0
	4 WL-X	-0.6	-1.0	0.0
	5 WL+Z	0.0	3.3	1.2
	6 WL-Z	0.0	-3.3	-1.2
2	1 DL	0.0	-0.9	0.0
	2 LL	0.9	12.9	0.0
	3 WL+X	0.6	1.0	0.0
	4 WL-X	-0.6	-1.0	0.0
	5 WL+Z	0.0	-3.3	0.0
	6 WL-Z	0.0	3.3	0.0
3	1 DL	0.0	-0.9	0.0
	2 LL	0.0	-14.1	-3.0
	3 WL+X	0.0	-1.0	0.0
	4 WL-X	0.0	1.0	0.0
	5 WL+Z	0.0	3.2	1.2
	6 WL-Z	0.0	-3.2	-1.2
4	1 DL	0.0	-0.9	0.0
	2 LL	0.0	6.5	0.0
	3 WL+X	0.0	-1.0	0.0
	4 WL-X	0.0	1.0	0.0
	5 WL+Z	0.0	-3.2	0.0
	6 WL-Z	0.0	3.2	0.0

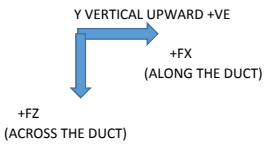




TW-1A

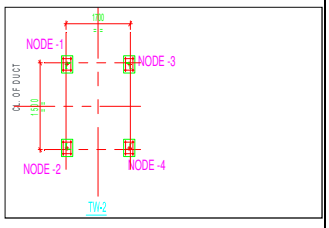
Node	L/C	Horizontal Fx Mton	Vertical Fy Mton	Horizontal Fz Mton
1	1 DL	0.0	-1.7	0.0
	2 LL	0.0	-5.8	-6.9
	3 WL+X	0.0	1.0	0.0
	4 WL-X	0.0	-1.0	0.0
	5 WL+Z	0.0	4.2	1.7
	6 WL-Z	0.0	-4.2	-1.7
2	1 DL	0.0	-1.7	0.0
	2 LL	0.0	7.1	0.0
	3 WL+X	0.0	1.0	0.0
	4 WL-X	0.0	-1.0	0.0
	5 WL+Z	0.0	-4.2	0.0
	6 WL-Z	0.0	4.2	0.0
3	1 DL	0.0	0.1	0.0
	2 LL	5.6	-8.2	3.1
	3 WL+X	0.6	-1.0	0.0
	4 WL-X	-0.6	1.0	0.0
	5 WL+Z	-0.2	4.5	1.4
	6 WL-Z	0.2	-4.5	-1.4
4	1 DL	0.0	0.04	0.0
	2 LL	-4.5	5.6	0.0
	3 WL+X	0.6	-1.0	0.0
	4 WL-X	-0.6	1.0	0.0
	5 WL+Z	0.2	-4.5	0.0
	6 WL-Z	-0.2	4.5	0.0





TW-2

Node	L/C	Horizontal Fx Mton	Vertical Fy Mton	Horizontal Fz Mton
1	1 DL	0.00	-0.90	0.00
	2 LL	0.74	7.55	1.73
	3 WL+X	0.56	0.88	0.00
	4 WL-X	-0.56	-0.88	0.00
	5 WL+Z	0.01	3.28	1.21
	6 WL-Z	-0.01	-3.28	-1.21
2	1 DL	0.00	-0.91	0.00
	2 LL	-1.41	-10.93	-0.01
	3 WL+X	0.55	0.86	0.00
	4 WL-X	-0.55	-0.86	0.00
	5 WL+Z	-0.01	-3.28	0.01
	6 WL-Z	0.01	3.28	-0.01
3	1 DL	0.00	-0.73	0.00
	2 LL	0.00	10.64	3.92
	3 WL+X	0.01	-0.88	0.00
	4 WL-X	-0.01	0.88	0.00
	5 WL+Z	0.00	3.19	1.20
	6 WL-Z	0.00	-3.19	-1.20
4	1 DL	0.00	-0.72	0.00
	2 LL	0.00	-9.90	-0.01
	3 WL+X	0.02	-0.86	0.00
	4 WL-X	-0.02	0.86	0.00
	5 WL+Z	0.00	-3.19	0.01
	6 WL-Z	0.00	3.19	-0.01

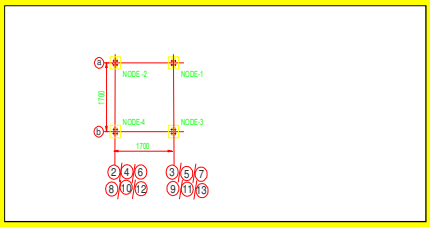


C01

LOAD DATA : POWER PLANT DUCT+PIPE+CABLE SUPPORTING TOWER

4L TOWER

Node	L/C	Horizontal Fx Mton	Vertical Fy Mton	Horizontal Fz Mton
1	1 DL	0.0	-3.0	0.0
	2 LL	0.7	-10.2	-4.0
	3 WL+X	1.2	2.0	0.0
	4 WL-X	-1.2	-2.0	0.0
	5 WL+Z	0.1	2.8	1.2
	6 WL-Z	-0.1	-2.8	-1.2
	10 LL1	1.7	-6.8	-2.9
2	1 DL	-0.1	-2.5	0.0
	2 LL	1.1	10.7	0.0
	3 WL+X	1.1	2.0	0.0
	4 WL-X	-1.1	-2.0	0.0
	5 WL+Z	-0.1	-2.8	0.0
	6 WL-Z	0.1	2.8	0.0
	10 LL1	2.1	10.5	0.0
3	1 DL	0.0	-2.1	0.0
	2 LL	0.1	-17.6	-4.2
	3 WL+X	0.0	-2.0	0.0
	4 WL-X	0.0	2.0	0.0
	5 WL+Z	0.0	3.0	1.2
	6 WL-Z	0.0	-3.0	-1.2
	10 LL1	0.0	-17.5	-3.1
4	1 DL	0.0	-1.7	0.0
	2 LL	-0.1	5.8	0.0
	3 WL+X	0.0	-2.0	0.0
	4 WL-X	0.0	2.0	0.0
	5 WL+Z	0.0	-3.0	0.0
	6 WL-Z	0.0	3.0	0.0
	10 LL1	-0.1	2.4	0.0



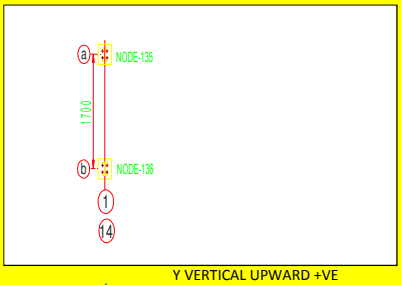
Y VERTICAL UPWARD +VE

+FZ

+FX

2L TRESTLE ALONG GRID 1 AND 14.

Node	L/C	Horizontal Fx Mton	Vertical Fy Mton	Horizontal Fz Mton
135	1 DL	0.0	-2.3	0.0
	2 LL	0.1	-1.2	0.0
	3 WL+X	1.1	1.8	0.0
	4 WL-X	-1.1	-1.8	0.0
	5 WL+Z	0.0	0.1	0.1
	6 WL-Z	0.0	-0.1	-0.1
	10 LL1	0.5	0.1	0.0
136	1 DL	0.0	-0.4	0.0
	2 LL	0.0	-1.5	0.0
1	3 WL+X	0.0	-1.8	0.0
	4 WL-X	0.0	1.8	0.1
	5 WL+Z	0.0	-0.1	-0.1
	6 WL-Z	0.0	0.1	0.0
	10 LL1	0.0	-1.4	0.0



Y VERTICAL UPWARD +VE

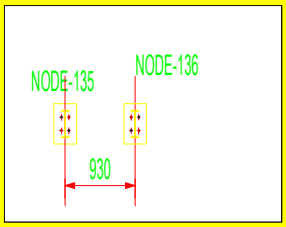
+FZ

+FX

2L TRESTLE ON SECTION G-G

Node	L/C	Horizontal Fx Mton	Vertical Fy Mton	Horizontal Fz Mton
135	1 DL	0.00	-1.99	0.00
	2 LL	-0.11	-3.95	0.00
	3 WL+X	0.02	1.04	0.00
	4 WL-X	-0.02	-1.04	0.00
	5 WL+Z	0.00	0.00	0.00
	6 WL-Z	0.00	0.00	0.00
	10 LL1	-0.04	0.24	0.00
136	1 DL	0.00	-0.54	0.00
	2 LL	0.11	-1.38	0.00
	3 WL+X	0.41	-1.04	0.00
	4 WL-X	-0.41	1.04	0.00
	5 WL+Z	0.00	0.00	0.00
	6 WL-Z	0.00	0.00	0.00
	10 LL1	1.48	-5.57	0.00

V



Y VERTICAL UPWARD +VE

+FX

+FZ

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

ATTACHMENT 2 STAAD Input & Output File of Foundation Mkd. F1

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUN  3, 2020                *
*           Time=    15:33:43                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: TOWER 1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 0; 14 0 -4 -0.25; 16 0 -4 1; 20 1 -4 1; 21 0.75 -4 1; 22 0.75 -4 0.75
     9. 23 1 -4 0.75; 24 0.5 -4 1; 25 0.5 -4 0.75; 26 0.25 -4 1; 27 0.25 -4 0.75
    10. 28 0 -4 0.75; 29 0.75 -4 0.5; 30 1 -4 0.5; 31 0.5 -4 0.5; 32 0.25 -4 0.5
    11. 33 0 -4 0.5; 34 0.75 -4 0.25; 35 1 -4 0.25; 36 0.5 -4 0.25; 37 0.25 -4 0.25
    12. 38 0 -4 0.25; 39 0.75 -4 0; 40 1 -4 0; 41 0.5 -4 0; 42 0.25 -4 0
    13. 43 0.75 -4 -0.25; 44 1 -4 -0.25; 45 0.5 -4 -0.25; 46 0.25 -4 -0.25
    14. 72 0.75 -4 -0.5; 73 1 -4 -0.5; 74 0.5 -4 -0.5; 75 0.25 -4 -0.5; 76 0 -4 -0.5
    15. 77 0.75 -4 -0.75; 78 1 -4 -0.75; 79 0.5 -4 -0.75; 80 0.25 -4 -0.75
    16. 81 0 -4 -0.75; 82 0.75 -4 -1; 83 1 -4 -1; 84 0.5 -4 -1; 85 0.25 -4 -1.
    17. 86 0 -4 -1; 90 -0.25 -4 1; 91 -0.25 -4 0.75; 92 -0.375 -4 1; 93 -0.375 -4 0.75
    18. 97 -0.25 -4 0.5; 98 -0.375 -4 0.5; 101 -0.25 -4 0.25; 102 -0.375 -4 0.25
    19. 105 -0.25 -4 0; 106 -0.375 -4 0; 109 -0.25 -4 -0.25; 110 -0.375 -4 -0.25
    20. 133 -0.25 -4 -0.5; 134 -0.375 -4 -0.5; 137 -0.25 -4 -0.75; 138 -0.375 -4 -0.75
    21. 141 -0.25 -4 -1; 142 -0.375 -4 -1; 145 0 -1 -0.75; 146 0 -1 0.75
    22. 147 0.75 -4 -1.25; 148 1 -4 -1.25; 149 0.5 -4 -1.25; 150 0.25 -4 -1.25
    23. 151 0 -4 -1.25; 152 -0.25 -4 -1.25; 153 -0.375 -4 -1.25; 156 1 -4 1.25
    24. 157 0.75 -4 1.25; 158 0.5 -4 1.25; 159 0.25 -4 1.25; 160 0 -4 1.25
    25. 161 -0.25 -4 1.25; 162 -0.375 -4 1.25; 181 1.5 -4 1; 182 1.5 -4 0.75
    26. 183 1.5 -4 0.5; 184 1.5 -4 0.25; 185 1.5 -4 0; 186 1.5 -4 -0.25
    27. 192 1.5 -4 -0.5; 193 1.5 -4 -0.75; 194 1.5 -4 -1; 195 1.5 -4 -1.25
    28. 196 1.5 -4 1.25; 220 1.5 -1 -0.75; 222 1.5 -1 0.75; 361 1.75 -4 1
    29. 362 1.75 -4 0.75; 363 1.75 -4 0.5; 364 1.75 -4 0.25; 365 1.75 -4 0
    30. 366 1.75 -4 -0.25; 372 1.75 -4 -0.5; 373 1.75 -4 -0.75; 374 1.75 -4 -1.
    31. 375 1.75 -4 -1.25; 376 1.75 -4 1.25; 419 1.875 -4 1; 420 1.875 -4 0.75
    32. 421 1.875 -4 0.5; 422 1.875 -4 0.25; 423 1.875 -4 0; 424 1.875 -4 -0.25
    33. 430 1.875 -4 -0.5; 431 1.875 -4 -0.75; 432 1.875 -4 -1; 433 1.875 -4 -1.25
    34. 434 1.875 -4 1.25; 435 0.75 -4 -1.5; 436 1 -4 -1.5; 437 0.5 -4 -1.5
    35. 438 0.25 -4 -1.5; 439 0 -4 -1.5; 440 -0.25 -4 -1.5; 441 -0.375 -4 -1.5
    36. 442 1.5 -4 -1.5; 451 1.75 -4 -1.5; 452 1.875 -4 -1.5; 453 1 -4 1.5
    37. 454 0.75 -4 1.5; 455 0.5 -4 1.5; 456 0.25 -4 1.5; 457 0 -4 1.5
    38. 458 -0.25 -4 1.5; 459 -0.375 -4 1.5; 460 1.5 -4 1.5; 469 1.75 -4 1.5
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39. 470 1.875 -4 1.5; 507 1.25 -4 1; 508 1.25 -4 0.75; 509 1.25 -4 0.5
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41. 514 1.25 -4 -0.75; 515 1.25 -4 -1; 516 1.25 -4 -1.25; 517 1.25 -4 1.25
42. 518 1.25 -4 -1.5; 519 1.25 -4 1.5; 572 0.75 -4 -1.75; 573 1 -4 -1.75
43. 574 0.5 -4 -1.75; 575 0.25 -4 -1.75; 576 0 -4 -1.75; 577 -0.25 -4 -1.75
44. 578 -0.375 -4 -1.75; 579 1.5 -4 -1.75; 580 1.75 -4 -1.75; 581 1.875 -4 -1.75
45. 596 1.25 -4 -1.75; 606 1 -4 1.75; 607 0.75 -4 1.75; 608 0.5 -4 1.75
46. 609 0.25 -4 1.75; 610 0 -4 1.75; 611 -0.25 -4 1.75; 612 -0.375 -4 1.75
47. 613 1.5 -4 1.75; 614 1.75 -4 1.75; 615 1.875 -4 1.75; 630 1.25 -4 1.75
48. MEMBER INCIDENCES
49. 133 81 145; 134 28 146; 201 193 220; 202 182 222
50. ELEMENT INCIDENCES SHELL
51. 25 20 21 22 23; 26 21 24 25 22; 27 24 26 27 25; 28 26 16 28 27; 29 23 22 29 30
52. 30 22 25 31 29; 31 25 27 32 31; 32 27 28 33 32; 33 30 29 34 35; 34 29 31 36 34
53. 35 31 32 37 36; 36 32 33 38 37; 37 35 34 39 40; 38 34 36 41 39; 39 36 37 42 41
54. 40 37 38 13 42; 41 40 39 43 44; 42 39 41 45 43; 43 41 42 46 45; 44 42 13 14 46
55. 65 44 43 72 73; 66 43 45 74 72; 67 45 46 75 74; 68 46 14 76 75; 69 73 72 77 78
56. 70 72 74 79 77; 71 74 75 80 79; 72 75 76 81 80; 73 78 77 82 83; 74 77 79 84 82
57. 75 79 80 85 84; 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91; 85 28 91 97 33
58. 86 91 93 98 97; 89 33 97 101 38; 90 97 98 102 101; 93 38 101 105 13
59. 94 101 102 106 105; 97 13 105 109 14; 98 105 106 110 109; 121 14 109 133 76
60. 122 109 110 134 133; 125 76 133 137 81; 126 133 134 138 137; 129 81 137 141 86
61. 130 137 138 142 141; 135 83 82 147 148; 136 82 84 149 147; 137 84 85 150 149
62. 138 85 86 151 150; 139 86 141 152 151; 140 141 142 153 152; 143 156 157 21 20
63. 144 157 158 24 21; 145 158 159 26 24; 146 159 160 16 26; 147 160 161 90 16
64. 148 161 162 92 90; 335 361 181 182 362; 336 362 182 183 363
65. 337 363 183 184 364; 338 364 184 185 365; 339 365 185 186 366
66. 345 366 186 192 372; 346 372 192 193 373; 347 373 193 194 374
67. 348 374 194 195 375; 349 376 196 181 361; 375 419 361 362 420
68. 376 420 362 363 421; 378 422 364 365 423; 379 423 365 366 424
69. 382 421 363 364 422; 385 424 366 372 430; 386 430 372 373 431
70. 387 431 373 374 432; 388 432 374 375 433; 389 434 376 361 419
71. 390 148 147 435 436; 391 147 149 437 435; 392 149 150 438 437
72. 393 150 151 439 438; 394 151 152 440 439; 395 152 153 441 440
73. 405 375 195 442 451; 406 433 375 451 452; 407 453 454 157 156
74. 408 454 455 158 157; 409 455 456 159 158; 410 456 457 160 159
75. 411 457 458 161 160; 412 458 459 162 161; 422 469 460 196 376
76. 423 470 469 376 434; 458 507 20 23 508; 459 508 23 30 509; 460 509 30 35 510
77. 461 510 35 40 511; 462 511 40 44 512; 464 512 44 73 513; 465 513 73 78 514
78. 466 514 78 83 515; 467 515 83 148 516; 468 517 156 20 507; 469 516 148 436 518
79. 470 519 453 156 517; 523 436 435 572 573; 524 435 437 574 572
80. 525 437 438 575 574; 526 438 439 576 575; 527 439 440 577 576
81. 528 440 441 578 577; 530 451 442 579 580; 531 452 451 580 581
82. 545 518 436 573 596; 555 606 607 454 453; 556 607 608 455 454
83. 557 608 609 456 455; 558 609 610 457 456; 559 610 611 458 457
84. 560 611 612 459 458; 562 614 613 460 469; 563 615 614 469 470
85. 577 630 606 453 519; 587 181 507 508 182; 588 182 508 509 183
86. 589 183 509 510 184; 590 184 510 511 185; 591 185 511 512 186
87. 593 186 512 513 192; 594 192 513 514 193; 595 193 514 515 194
88. 596 194 515 516 195; 597 196 517 507 181; 598 195 516 518 442
89. 599 460 519 517 196; 600 442 518 596 579; 602 613 630 519 460
90. START GROUP DEFINITION
91. ELEMENT
92. _MOMENT 25 TO 27 29 TO 31 33 TO 44 65 TO 71 73 TO 75 82 86 89 90 93 94 97 -
93. 98 121 122 126 130 135 TO 140 143 TO 148 337 TO 339 345 348 349 375 376 378 -
94. 379 382 385 TO 395 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 -
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 95. 531 545 555 TO 560 562 563 577 589 TO 591 593 596 TO 600 602
 96. _SHEAR 25 26 29 30 33 34 37 TO 44 65 66 69 70 73 74 93 94 97 98 135 136 143 -
 97. 144 338 339 378 379 390 TO 395 405 TO 412 422 423 461 462 469 470 -
 98. 523 TO 528 530 531 545 555 TO 560 562 563 577 590 591 598 TO 600 602
 99. END GROUP DEFINITION
100. ELEMENT PROPERTY
101. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
102. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
103. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
104. 562 563 577 587 TO 591 593 TO 600 602 THICKNESS 0.4
105. SUPPORTS
106. 13 14 16 20 TO 46 72 TO 86 90 TO 93 97 98 101 102 105 106 109 110 133 134 -
107. 137 138 141 142 147 TO 153 156 TO 162 181 TO 186 192 TO 196 361 TO 366 372 -
108. 373 TO 376 419 TO 424 430 TO 442 451 TO 460 469 470 507 TO 519 572 TO 581 -
109. 596 606 TO 615 630 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT COMPRESSION
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16	0.062500
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43	0.062500
44	0.062500
45	0.062500
46	0.062500
72	0.062500
73	0.062500
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500

79	0.062500
80	0.062500
81	0.062500
82	0.062500
83	0.062500
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91	0.046875
92	0.015625
93	0.015625
97	0.046875
98	0.015625
101	0.046875
102	0.015625
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106	0.015625
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110	0.015625
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134	0.015625
137	0.046875
138	0.015625
141	0.046875
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150	0.062500
151	0.062500
152	0.046875
153	0.015625
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196	0.062500
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366	0.046875
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376	0.046875
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575	0.031250
576	0.031250
577	0.023437

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       579          0.031250
       580          0.023437
       581          0.007813
       596          0.031250
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       608          0.031250
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       610          0.031250
       611          0.023437
       612          0.007812
       613          0.031250
       614          0.023437
       615          0.007813
       630          0.031250
    TOTAL           7.875000
110. DEFINE MATERIAL START
111. ISOTROPIC STEEL
112. E 2.09042E+007
113. POISSON 0.3
114. DENSITY 7.83341
115. ALPHA 1.2E-005
116. DAMP 0.03
117. TYPE STEEL
118. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
119. ISOTROPIC CONCRETE
120. E 2.21467E+006
121. POISSON 0.17
122. DENSITY 2.40262
123. ALPHA 1E-005
124. DAMP 0.05
125. TYPE CONCRETE
126. STRENGTH FCU 2812.28
127. END DEFINE MATERIAL
128. MEMBER PROPERTY AMERICAN
129. 133 134 201 202 PRIS YD 0.55 ZD 0.55
130. CONSTANTS
131. MATERIAL CONCRETE ALL
132. ************************************************************************
133. LOAD 1 DL+LL
134. SELFWEIGHT Y -1
135. ELEMENT LOAD
136. *SOIL+SURCHARGE=2.2X1.8+1=5T/SQ.M)
137. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
138. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
139. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
140. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5
141. JOINT LOAD
142. 145 FX 0.88 FY -9.01 FZ -3.03
143. 220 FX 0 FY -15.02 FZ -3
144. 146 FX 0.92 FY 12.02 FZ 0.01
145. 222 FX 0 FY 5.55 FZ 0.02
146. ************************************************************************8
147. LOAD 2 DL+0.75*LL+0.75*WL1
148. SELFWEIGHT Y -1
```

```
149. ELEMENT LOAD
150. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
151. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
152. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
153. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5
154. JOINT LOAD
155. 145 FX 1.08 FY -6.24 FZ -2.27
156. 220 FX 0.01 FY -12.25 FZ -2.25
157. 146 FX 1.11 FY 9.51 FZ 0.01
158. 222 FX 0.01 FY 3.2 FZ 0.02
159. ********************************************************************************8
160. LOAD 3 DL+0.75*LL+0.75*WL2
161. SELFWEIGHT Y -1
162. ELEMENT LOAD
163. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
164. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
165. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
166. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5
167. JOINT LOAD
168. 145 FX 0.24 FY -7.74 FZ -2.27
169. 220 FX -0.01 FY -10.75 FZ -2.25
170. 146 FX 0.28 FY 8.05 FZ 0.01
171. 222 FX -0.02 FY 4.66 FZ 0.01
172. ********************************************************************************
173. LOAD 4 DL+0.75*LL+0.75*WL3
174. SELFWEIGHT Y -1
175. ELEMENT LOAD
176. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
177. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
178. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
179. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5
180. JOINT LOAD
181. 145 FX 0.67 FY -4.52 FZ -1.36
182. 220 FX 0.69 FY 6.32 FZ 0.02
183. 146 FX 0 FY -9.11 FZ -1.35
184. 222 FX 0 FY 1.54 FZ 0.02
185. ********************************************************************************
186. LOAD 5 DL+0.75*LL+0.75*WL4
187. SELFWEIGHT Y -1
188. ELEMENT LOAD
189. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
190. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
191. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
192. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5
193. JOINT LOAD
194. 145 FX 0.65 FY -9.45 FZ -3.18
195. 220 FX 0 FY -13.89 FZ -3.15
196. 146 FX 0.7 FY 11.24 FZ 0
197. 222 FX 0 FY 6.32 FZ 0.01
198. ********************************************************************************
199. LOAD 6 0.6*DL+WL1
200. SELFWEIGHT Y -1
201. ELEMENT LOAD
202. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
203. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
204. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
```

```
205. 562 563 577 587 TO 591 593 TO 600 602 PR GY -3.4
206. JOINT LOAD
207. 145 FX 0.56 FY 0.45 FZ 0
208. 220 FX 0.01 FY -1.56 FZ 0
209. 146 FX 0.55 FY 0.42 FZ 0
210. 222 FX 0.02 FY -1.53 FZ 0
211. ********************************************************************************
212. LOAD 7 0.6*DL+WL2
213. SELFWEIGHT Y -1
214. ELEMENT LOAD
215. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
216. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
217. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
218. 562 563 577 587 TO 591 593 TO 600 602 PR GY -3.4
219. JOINT LOAD
220. 145 FX -0.56 FY -1.56 FZ 0
221. 220 FX -0.01 FY 0.45 FZ 0
222. 146 FX -0.55 FY -1.53 FZ 0
223. 222 FX -0.02 FY 0.42 FZ 0
224. ********************************************************************************
225. LOAD 8 0.6*DL+WL3
226. SELFWEIGHT Y -1
227. ELEMENT LOAD
228. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
229. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
230. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
231. 562 563 577 587 TO 591 593 TO 600 602 PR GY -3.4
232. JOINT LOAD
233. 145 FX 0.01 FY 2.73 FZ 1.21
234. 220 FX 0 FY 2.63 FZ 1.2
235. 146 FX -0.01 FY -3.84 FZ 0.01
236. 222 FX 0 FY -3.74 FZ 0.01
237. ********************************************************************************
238. LOAD 9 0.6*DL+WL4
239. SELFWEIGHT Y -1
240. ELEMENT LOAD
241. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
242. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
243. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
244. 562 563 577 587 TO 591 593 TO 600 602 PR GY -3.4
245. JOINT LOAD
246. 145 FX -0.01 FY -3.84 FZ -1.21
247. 220 FX 0 FY -3.74 FZ -1.2
248. 146 FX 0.01 FY 2.73 FZ -0.01
249. 222 FX 0 FY 2.63 FZ -0.01
250. ********************************************************************888
251. ********************************************************************************
252. LOAD 100 1.4D
253. *SOIL+SURCHARGE=2.2X1.8*1.4+1=6.5T/SQ.M)
254. SELFWEIGHT Y -1
255. ELEMENT LOAD
256. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
257. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
258. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
259. 562 563 577 587 TO 591 593 TO 600 602 PR GY -6.5
260. JOINT LOAD
```

```
261. 145 FX 0 FY -1.3 FZ 0
262. 220 FX 0 FY -1.3 FZ 0
263. 146 FX 0 FY -1.3 FZ 0
264. 222 FX 0 FY -1.3 FZ 0
265. ************************************************************************
266. LOAD 101 1.4D+1.7(LL)
267. SELFWEIGHT Y -1
268. ELEMENT LOAD
269. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
270. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
271. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
272. 562 563 577 587 TO 591 593 TO 600 602 PR GY -6.5
273. JOINT LOAD
274. 145 FX 1.5 FY -15.04 FZ -5.15
275. 220 FX 0 FY -25.26 FZ -5.1
276. 146 FX 1.57 FY 20.71 FZ 0.02
277. 222 FX -0.01 FY 9.71 FZ 0.03
278. ************************************************************************
279. LOAD 102 1.2D+1.0(LL)+1.6WL(+X)
280. SELFWEIGHT Y -1
281. ELEMENT LOAD
282. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
283. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
284. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
285. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5.8
286. JOINT LOAD
287. 145 FX 1.77 FY -7.59 FZ -3.03
288. 220 FX 0.03 FY -16.81 FZ -3.01
289. 146 FX 1.81 FY 13.39 FZ 0.01
290. 222 FX 0.02 FY 3.81 FZ 0.02
291. ************************************************************************
292. LOAD 103 1.2D+1.0(LL)+1.6WL(-X)
293. SELFWEIGHT Y -1
294. ELEMENT LOAD
295. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
296. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
297. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
298. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5.8
299. JOINT LOAD
300. 145 FX -0.01 FY -10.8 FZ -3.03
301. 220 FX -0.02 FY -13.6 FZ -3
302. 146 FX 0.04 FY 10.27 FZ 0.01
303. 222 FX -0.03 FY 6.93 FZ 0.02
304. ************************************************************************
305. LOAD 104 1.2D+1.0(LL)+1.6WL(+Z)
306. SELFWEIGHT Y -1
307. ELEMENT LOAD
308. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
309. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
310. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
311. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5.8
312. JOINT LOAD
313. 145 FX 0.89 FY -3.94 FZ -1.09
314. 220 FX 0 FY -10.11 FZ -1.08
315. 146 FX 0.91 FY 6.57 FZ 0.03
316. 222 FX 0 FY 0.27 FZ 0.04
```

```
317. ********************************************************************************
318. LOAD 105 1.2D+1.0(LL)+1.6WL(-Z)
319. SELFWEIGHT Y -1
320. ELEMENT LOAD
321. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
322. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
323. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
324. 562 563 577 587 TO 591 593 TO 600 602 PR GY -5.8
325. JOINT LOAD
326. 145 FX 0.87 FY -14.45 FZ -4.97
327. 220 FX 0 FY -20.31 FZ -4.92
328. 146 FX 0.94 FY 17.09 FZ -0.01
329. 222 FX -0.01 FY 10.46 FZ 0
330. ********************************************************************************
331. LOAD 106 0.9*D+1.6WL(+X)
332. SELFWEIGHT Y -1
333. ELEMENT LOAD
334. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
335. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
336. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
337. 562 563 577 587 TO 591 593 TO 600 602 PR GY -4.6
338. JOINT LOAD
339. 145 FX 0.89 FY 0.77 FZ 0
340. 220 FX 0.02 FY -2.44 FZ 0
341. 146 FX 0.89 FY 0.72 FZ 0
342. 222 FX 0.02 FY -2.39 FZ 0
343. ********************************************************************************
344. LOAD 107 0.9*D+1.6WL(-X)
345. SELFWEIGHT Y -1
346. ELEMENT LOAD
347. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
348. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
349. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
350. 562 563 577 587 TO 591 593 TO 600 602 PR GY -4.6
351. JOINT LOAD
352. 145 FX -0.89 FY -2.44 FZ 0
353. 220 FX -0.02 FY 0.77 FZ 0
354. 146 FX -0.89 FY -2.39 FZ 0
355. 222 FX -0.02 FY 0.73 FZ 0
356. ********************************************************************************
357. LOAD 108 0.9*D+1.6WL(+Z)
358. SELFWEIGHT Y -1
359. ELEMENT LOAD
360. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
361. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
362. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
363. 562 563 577 587 TO 591 593 TO 600 602 PR GY -4.6
364. JOINT LOAD
365. 145 FX 0.01 FY 4.42 FZ 1.94
366. 220 FX -0.01 FY -6.09 FZ 0.02
367. 146 FX 0 FY 4.26 FZ 1.92
368. 222 FX 0 FY -5.93 FZ 0.02
369. ********************************************************************************
370. LOAD 109 0.9*D+1.6WL(-Z)
371. SELFWEIGHT Y -1
372. ELEMENT LOAD
```

```
373. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
374. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
375. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
376. 562 563 577 587 TO 591 593 TO 600 602 PR GY -4.6
377. JOINT LOAD
378. 145 FX -0.01 FY -6.09 FZ -1.94
379. 220 FX 0 FY -5.93 FZ -1.92
380. 146 FX 0.01 FY 4.42 FZ -0.02
381. 222 FX 0 FY 4.26 FZ -0.02
382. ******************************************************************************
383. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**169**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**140**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**165**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 141/ 19/ 66 DOF
TOTAL PRIMARY LOAD CASES = 19, TOTAL DEGREES OF FREEDOM = 519
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 35 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.2/ 164306.0 MB

```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=    1


**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=    2


**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=    3



**NOTE-Tension/Compression converged after  1 iterations, Case=    4
```

```
**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=    5



**NOTE-Tension/Compression converged after  1 iterations, Case=    6



**NOTE-Tension/Compression converged after  1 iterations, Case=    7



**NOTE-Tension/Compression converged after  1 iterations, Case=    8



**NOTE-Tension/Compression converged after  1 iterations, Case=    9



**NOTE-Tension/Compression converged after  1 iterations, Case=  100


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=  101


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4

**NOTE-Tension/Compression converged after  4 iterations, Case=  102


**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  103


**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  104


**START ITERATION NO.          2
**START ITERATION NO.          3
**START ITERATION NO.          4
```

**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

384. LOAD LIST 100 TO 109
385. START CONCRETE DESIGN

```
386. CODE ACI 1999
387. TRACK 0 MEMB 133 134
388. MAXMAIN 25 MEMB 133 134
389. CLB 0.075 MEMB 133 134
390. CLS 0.075 MEMB 133 134
391. DESIGN COLUMN 133 134
```

==

COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

392. END CONCRETE DESIGN
393. FINISH

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

ATTACHMENT 3 STAAD Input & Output File of Foundation Mkd. F2

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUN  3, 2020                *
*           Time=    14:45:35                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: TOWER 2.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 0; 14 0 -4 -0.25; 16 0 -4 1; 20 1 -4 1; 21 0.75 -4 1; 22 0.75 -4 0.75
     9. 23 1 -4 0.75; 24 0.5 -4 1; 25 0.5 -4 0.75; 26 0.25 -4 1; 27 0.25 -4 0.75
    10. 28 0 -4 0.75; 29 0.75 -4 0.5; 30 1 -4 0.5; 31 0.5 -4 0.5; 32 0.25 -4 0.5
    11. 33 0 -4 0.5; 34 0.75 -4 0.25; 35 1 -4 0.25; 36 0.5 -4 0.25; 37 0.25 -4 0.25
    12. 38 0 -4 0.25; 39 0.75 -4 0; 40 1 -4 0; 41 0.5 -4 0; 42 0.25 -4 0
    13. 43 0.75 -4 -0.25; 44 1 -4 -0.25; 45 0.5 -4 -0.25; 46 0.25 -4 -0.25
    14. 72 0.75 -4 -0.5; 73 1 -4 -0.5; 74 0.5 -4 -0.5; 75 0.25 -4 -0.5; 76 0 -4 -0.5
    15. 77 0.75 -4 -0.75; 78 1 -4 -0.75; 79 0.5 -4 -0.75; 80 0.25 -4 -0.75
    16. 81 0 -4 -0.75; 82 0.75 -4 -1; 83 1 -4 -1; 84 0.5 -4 -1; 85 0.25 -4 -1.
    17. 86 0 -4 -1; 90 -0.25 -4 1; 91 -0.25 -4 0.75; 92 -0.4 -4 1; 93 -0.4 -4 0.75
    18. 97 -0.25 -4 0.5; 98 -0.4 -4 0.5; 101 -0.25 -4 0.25; 102 -0.4 -4 0.25
    19. 105 -0.25 -4 0; 106 -0.4 -4 0; 109 -0.25 -4 -0.25; 110 -0.4 -4 -0.25
    20. 133 -0.25 -4 -0.5; 134 -0.4 -4 -0.5; 137 -0.25 -4 -0.75; 138 -0.4 -4 -0.75
    21. 141 -0.25 -4 -1; 142 -0.4 -4 -1; 145 0 -1 -0.75; 146 0 -1 0.75
    22. 147 0.75 -4 -1.25; 148 1 -4 -1.25; 149 0.5 -4 -1.25; 150 0.25 -4 -1.25
    23. 151 0 -4 -1.25; 152 -0.25 -4 -1.25; 153 -0.4 -4 -1.25; 156 1 -4 1.25
    24. 157 0.75 -4 1.25; 158 0.5 -4 1.25; 159 0.25 -4 1.25; 160 0 -4 1.25
    25. 161 -0.25 -4 1.25; 162 -0.4 -4 1.25; 181 1.5 -4 1; 182 1.5 -4 0.75
    26. 183 1.5 -4 0.5; 184 1.5 -4 0.25; 185 1.5 -4 0; 186 1.5 -4 -0.25
    27. 192 1.5 -4 -0.5; 193 1.5 -4 -0.75; 194 1.5 -4 -1; 195 1.5 -4 -1.25
    28. 196 1.5 -4 1.25; 220 1.7 -1 -0.75; 222 1.7 -1 0.75; 361 1.7 -4 1
    29. 362 1.7 -4 0.75; 363 1.7 -4 0.5; 364 1.7 -4 0.25; 365 1.7 -4 0
    30. 366 1.7 -4 -0.25; 372 1.7 -4 -0.5; 373 1.7 -4 -0.75; 374 1.7 -4 -1.
    31. 375 1.7 -4 -1.25; 376 1.7 -4 1.25; 419 1.95 -4 1; 420 1.95 -4 0.75
    32. 421 1.95 -4 0.5; 422 1.95 -4 0.25; 423 1.95 -4 0; 424 1.95 -4 -0.25
    33. 430 1.95 -4 -0.5; 431 1.95 -4 -0.75; 432 1.95 -4 -1; 433 1.95 -4 -1.25
    34. 434 1.95 -4 1.25; 435 0.75 -4 -1.5; 436 1 -4 -1.5; 437 0.5 -4 -1.5
    35. 438 0.25 -4 -1.5; 439 0 -4 -1.5; 440 -0.25 -4 -1.5; 441 -0.4 -4 -1.5
    36. 442 1.5 -4 -1.5; 451 1.7 -4 -1.5; 452 1.95 -4 -1.5; 453 1 -4 1.5
    37. 454 0.75 -4 1.5; 455 0.5 -4 1.5; 456 0.25 -4 1.5; 457 0 -4 1.5
    38. 458 -0.25 -4 1.5; 459 -0.4 -4 1.5; 460 1.5 -4 1.5; 469 1.7 -4 1.5
```

```
39. 470 1.95 -4 1.5; 507 1.25 -4 1; 508 1.25 -4 0.75; 509 1.25 -4 0.5
40. 510 1.25 -4 0.25; 511 1.25 -4 0; 512 1.25 -4 -0.25; 513 1.25 -4 -0.5
41. 514 1.25 -4 -0.75; 515 1.25 -4 -1; 516 1.25 -4 -1.25; 517 1.25 -4 1.25
42. 518 1.25 -4 -1.5; 519 1.25 -4 1.5; 572 0.75 -4 -1.75; 573 1 -4 -1.75
43. 574 0.5 -4 -1.75; 575 0.25 -4 -1.75; 576 0 -4 -1.75; 577 -0.25 -4 -1.75
44. 578 -0.4 -4 -1.75; 579 1.5 -4 -1.75; 580 1.7 -4 -1.75; 581 1.95 -4 -1.75
45. 596 1.25 -4 -1.75; 606 1 -4 1.75; 607 0.75 -4 1.75; 608 0.5 -4 1.75
46. 609 0.25 -4 1.75; 610 0 -4 1.75; 611 -0.25 -4 1.75; 612 -0.4 -4 1.75
47. 613 1.5 -4 1.75; 614 1.7 -4 1.75; 615 1.95 -4 1.75; 630 1.25 -4 1.75
48. 631 2.1 -4 1; 632 2.1 -4 0.75; 633 2.1 -4 0.5; 634 2.1 -4 0.25; 635 2.1 -4 0
49. 636 2.1 -4 -0.25; 637 2.1 -4 -0.5; 638 2.1 -4 -0.75; 639 2.1 -4 -1.
50. 640 2.1 -4 -1.25; 641 2.1 -4 1.25; 642 2.1 -4 -1.5; 643 2.1 -4 1.5
51. 644 2.1 -4 -1.75; 645 2.1 -4 1.75
52. MEMBER INCIDENCES
53. 133 81 145; 134 28 146; 201 373 220; 202 362 222
54. ELEMENT INCIDENCES SHELL
55. 25 20 21 22 23; 26 21 24 25 22; 27 24 26 27 25; 28 26 16 28 27; 29 23 22 29 30
56. 30 22 25 31 29; 31 25 27 32 31; 32 27 28 33 32; 33 30 29 34 35; 34 29 31 36 34
57. 35 31 32 37 36; 36 32 33 38 37; 37 35 34 39 40; 38 34 36 41 39; 39 36 37 42 41
58. 40 37 38 13 42; 41 40 39 43 44; 42 39 41 45 43; 43 41 42 46 45; 44 42 13 14 46
59. 65 44 43 72 73; 66 43 45 74 72; 67 45 46 75 74; 68 46 14 76 75; 69 73 72 77 78
60. 70 72 74 79 77; 71 74 75 80 79; 72 75 76 81 80; 73 78 77 82 83; 74 77 79 84 82
61. 75 79 80 85 84; 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91; 85 28 91 97 33
62. 86 91 93 98 97; 89 33 97 101 38; 90 97 98 102 101; 93 38 101 105 13
63. 94 101 102 106 105; 97 13 105 109 14; 98 105 106 110 109; 121 14 109 133 76
64. 122 109 110 134 133; 125 76 133 137 81; 126 133 134 138 137; 129 81 137 141 86
65. 130 137 138 142 141; 135 83 82 147 148; 136 82 84 149 147; 137 84 85 150 149
66. 138 85 86 151 150; 139 86 141 152 151; 140 141 142 153 152; 143 156 157 21 20
67. 144 157 158 24 21; 145 158 159 26 24; 146 159 160 16 26; 147 160 161 90 16
68. 148 161 162 92 90; 335 361 181 182 362; 336 362 182 183 363
69. 337 363 183 184 364; 338 364 184 185 365; 339 365 185 186 366
70. 345 366 186 192 372; 346 372 192 193 373; 347 373 193 194 374
71. 348 374 194 195 375; 349 376 196 181 361; 375 419 361 362 420
72. 376 420 362 363 421; 378 422 364 365 423; 379 423 365 366 424
73. 382 421 363 364 422; 385 424 366 372 430; 386 430 372 373 431
74. 387 431 373 374 432; 388 432 374 375 433; 389 434 376 361 419
75. 390 148 147 435 436; 391 147 149 437 435; 392 149 150 438 437
76. 393 150 151 439 438; 394 151 152 440 439; 395 152 153 441 440
77. 405 375 195 442 451; 406 433 375 451 452; 407 453 454 157 156
78. 408 454 455 158 157; 409 455 456 159 158; 410 456 457 160 159
79. 411 457 458 161 160; 412 458 459 162 161; 422 469 460 196 376
80. 423 470 469 376 434; 458 507 20 23 508; 459 508 23 30 509; 460 509 30 35 510
81. 461 510 35 40 511; 462 511 40 44 512; 464 512 44 73 513; 465 513 73 78 514
82. 466 514 78 83 515; 467 515 83 148 516; 468 517 156 20 507; 469 516 148 436 518
83. 470 519 453 156 517; 523 436 435 572 573; 524 435 437 574 572
84. 525 437 438 575 574; 526 438 439 576 575; 527 439 440 577 576
85. 528 440 441 578 577; 530 451 442 579 580; 531 452 451 580 581
86. 545 518 436 573 596; 555 606 607 454 453; 556 607 608 455 454
87. 557 608 609 456 455; 558 609 610 457 456; 559 610 611 458 457
88. 560 611 612 459 458; 562 614 613 460 469; 563 615 614 469 470
89. 577 630 606 453 519; 587 181 507 508 182; 588 182 508 509 183
90. 589 183 509 510 184; 590 184 510 511 185; 591 185 511 512 186
91. 593 186 512 513 192; 594 192 513 514 193; 595 193 514 515 194
92. 596 194 515 516 195; 597 196 517 507 181; 598 195 516 518 442
93. 599 460 519 517 196; 600 442 518 596 579; 602 613 630 519 460
94. 603 631 419 420 632; 604 632 420 421 633; 605 633 421 422 634
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 95. 606 634 422 423 635; 607 635 423 424 636; 608 636 424 430 637
 96. 609 637 430 431 638; 610 638 431 432 639; 611 639 432 433 640
 97. 612 641 434 419 631; 613 640 433 452 642; 614 643 470 434 641
 98. 615 642 452 581 644; 616 645 615 470 643
 99. START GROUP DEFINITION
100. ELEMENT
101. _MOMENT 25 TO 27 29 TO 31 33 TO 44 65 TO 71 73 TO 75 82 86 89 90 93 94 97 -
102. 98 121 122 126 130 135 TO 140 143 TO 148 337 TO 339 345 348 349 378 379 382 -
103. 385 388 TO 395 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 -
104. 545 555 TO 560 562 563 577 587 TO 591 593 TO 600 602 TO 616
105. _SHEAR 25 26 29 30 33 34 37 TO 44 65 66 69 70 73 74 93 94 97 98 135 136 143 -
106. 144 338 339 378 379 390 TO 395 405 TO 412 422 423 458 TO 462 464 TO 470 523 -
107. 524 TO 528 530 531 545 555 TO 560 562 563 577 590 591 598 TO 600 602 606 607 -
108. 613 TO 616
109. END GROUP DEFINITION
110. ELEMENT PROPERTY
111. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
112. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
113. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
114. 562 563 577 587 TO 591 593 TO 600 602 TO 616 THICKNESS 0.4
115. SUPPORTS
116. 13 14 16 20 TO 46 72 TO 86 90 TO 93 97 98 101 102 105 106 109 110 133 134 -
117. 137 138 141 142 147 TO 153 156 TO 162 181 TO 186 192 TO 196 361 TO 366 372 -
118. 373 TO 376 419 TO 424 430 TO 442 451 TO 460 469 470 507 TO 519 572 TO 581 -
119. 596 606 TO 615 630 TO 645 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
```

JOINT	INFLUENCE AREA
13	0.062500
14	0.062500
16	0.062500
20	0.062500
21	0.062500
22	0.062500
23	0.062500
24	0.062500
25	0.062500
26	0.062500
27	0.062500
28	0.062500
29	0.062500
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.062500
36	0.062500
37	0.062500
38	0.062500
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500

44	0.062500
45	0.062500
46	0.062500
72	0.062500
73	0.062500
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062500
80	0.062500
81	0.062500
82	0.062500
83	0.062500
84	0.062500
85	0.062500
86	0.062500
90	0.050000
91	0.050000
92	0.018750
93	0.018750
97	0.050000
98	0.018750
101	0.050000
102	0.018750
105	0.050000
106	0.018750
109	0.050000
110	0.018750
133	0.050000
134	0.018750
137	0.050000
138	0.018750
141	0.050000
142	0.018750
147	0.062500
148	0.062500
149	0.062500
150	0.062500
151	0.062500
152	0.050000
153	0.018750
156	0.062500
157	0.062500
158	0.062500
159	0.062500
160	0.062500
161	0.050000
162	0.018750
181	0.056250
182	0.056250
183	0.056250
184	0.056250
185	0.056250
186	0.056250

192	0.056250
193	0.056250
194	0.056250
195	0.056250
196	0.056250
361	0.056250
362	0.056250
363	0.056250
364	0.056250
365	0.056250
366	0.056250
372	0.056250
373	0.056250
374	0.056250
375	0.056250
376	0.056250
419	0.050000
420	0.050000
421	0.050000
422	0.050000
423	0.050000
424	0.050000
430	0.050000
431	0.050000
432	0.050000
433	0.050000
434	0.050000
435	0.062500
436	0.062500
437	0.062500
438	0.062500
439	0.062500
440	0.050000
441	0.018750
442	0.056250
451	0.056250
452	0.050000
453	0.062500
454	0.062500
455	0.062500
456	0.062500
457	0.062500
458	0.050000
459	0.018750
460	0.056250
469	0.056250
470	0.050000
507	0.062500
508	0.062500
509	0.062500
510	0.062500
511	0.062500
512	0.062500
513	0.062500
514	0.062500
515	0.062500

516	0.062500
517	0.062500
518	0.062500
519	0.062500
572	0.031250
573	0.031250
574	0.031250
575	0.031250
576	0.031250
577	0.025000
578	0.009375
579	0.028125
580	0.028125
581	0.025000
596	0.031250
606	0.031250
607	0.031250
608	0.031250
609	0.031250
610	0.031250
611	0.025000
612	0.009375
613	0.028125
614	0.028125
615	0.025000
630	0.031250
631	0.018750
632	0.018750
633	0.018750
634	0.018750
635	0.018750
636	0.018750
637	0.018750
638	0.018750
639	0.018750
640	0.018750
641	0.018750
642	0.018750
643	0.018750
644	0.009375
645	0.009375
TOTAL	8.750004

```
120. DEFINE MATERIAL START
121. ISOTROPIC STEEL
122. E 2.09042E+007
123. POISSON 0.3
124. DENSITY 7.83341
125. ALPHA 1.2E-005
126. DAMP 0.03
127. TYPE STEEL
128. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
129. ISOTROPIC CONCRETE
130. E 2.21467E+006
131. POISSON 0.17
132. DENSITY 2.40262
133. ALPHA 1E-005
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134. DAMP 0.05
135. TYPE CONCRETE
136. STRENGTH FCU 2812.28
137. END DEFINE MATERIAL
138. MEMBER PROPERTY AMERICAN
139. 133 134 201 202 PRIS YD 0.55 ZD 0.55
140. CONSTANTS
141. MATERIAL CONCRETE ALL
142. ******************************************************************************
143. LOAD 1 DL+LL
144. SELFWEIGHT Y -1
145. ELEMENT LOAD
146. *SOIL+SURCHARGE=2.2X1.8+1=5 T/SQ.M)
147. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
148. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
149. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
150. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5
151. JOINT LOAD
152. 145 FX 0.74 FY 6.65 FZ 1.73
153. 220 FX 0 FY 9.91 FZ 3.92
154. 146 FX -1.41 FY -11.83 FZ -0.01
155. 222 FX 0 FY -10.63 FZ -0.01
156. ******************************************************************************8
157. LOAD 2 DL+0.75*LL+0.75*WL1
158. SELFWEIGHT Y -1
159. ELEMENT LOAD
160. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
161. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
162. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
163. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5
164. JOINT LOAD
165. 145 FX 0.97 FY 5.42 FZ 1.3
166. 220 FX 0.01 FY 6.59 FZ 2.94
167. 146 FX -0.65 FY -8.46 FZ -0.01
168. 222 FX 0.01 FY -8.8 FZ -0.01
169. ******************************************************************************8
170. LOAD 3 DL+0.75*LL+0.75*WL2
171. SELFWEIGHT Y -1
172. ELEMENT LOAD
173. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
174. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
175. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
176. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5
177. JOINT LOAD
178. 145 FX 0.14 FY 4.1 FZ 1.3
179. 220 FX -0.01 FY 7.92 FZ 2.94
180. 146 FX -1.48 FY -9.75 FZ -0.01
181. 222 FX -0.01 FY -7.5 FZ -0.01
182. ******************************************************************************
183. LOAD 4 DL+0.75*LL+0.75*WL3
184. SELFWEIGHT Y -1
185. ELEMENT LOAD
186. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
187. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
188. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
189. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5
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190. JOINT LOAD
191. 145 FX 0.56 FY 7.22 FZ 2.21
192. 220 FX 0 FY 9.64 FZ 3.84
193. 146 FX -1.07 FY -11.57 FZ 0
194. 222 FX 0 FY -10.54 FZ 0
195. ******************************************************************
196. LOAD 5 DL+0.75*LL+0.75*WL4
197. SELFWEIGHT Y -1
198. ELEMENT LOAD
199. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
200. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
201. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
202. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5
203. JOINT LOAD
204. 145 FX 0.55 FY 2.3 FZ 0.39
205. 220 FX 0 FY 4.86 FZ 2.04
206. 146 FX -1.05 FY -6.64 FZ -0.02
207. 222 FX 0 FY -5.76 FZ -0.02
208. ******************************************************************
209. LOAD 6 0.6*DL+WL1
210. SELFWEIGHT Y -1
211. ELEMENT LOAD
212. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
213. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
214. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
215. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -3.4
216. JOINT LOAD
217. 145 FX 0.56 FY 0.34 FZ 0
218. 220 FX 0.01 FY -1.32 FZ 0
219. 146 FX 0.55 FY 0.32 FZ 0
220. 222 FX 0.02 FY -1.29 FZ 0
221. ******************************************************************
222. LOAD 7 0.6*DL+WL2
223. SELFWEIGHT Y -1
224. ELEMENT LOAD
225. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
226. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
227. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
228. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -3.4
229. JOINT LOAD
230. 145 FX -0.56 FY -1.43 FZ 0
231. 220 FX -0.01 FY 0.45 FZ 0
232. 146 FX -0.55 FY -1.4 FZ 0
233. 222 FX -0.02 FY 0.43 FZ 0
234. ******************************************************************
235. LOAD 8 0.6*DL+WL3
236. SELFWEIGHT Y -1
237. ELEMENT LOAD
238. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
239. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
240. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
241. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -3.4
242. JOINT LOAD
243. 145 FX 0.01 FY 2.74 FZ 1.21
244. 220 FX 0 FY 2.75 FZ 1.2
245. 146 FX -0.01 FY -3.83 FZ 0.01
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246. 222 FX 0 FY -3.62 FZ 0.01
247. ********************************************************************************
248. LOAD 9 0.6*DL+WL4
249. SELFWEIGHT Y -1
250. ELEMENT LOAD
251. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
252. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
253. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
254. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -3.4
255. JOINT LOAD
256. 145 FX -0.01 FY -3.83 FZ -1.21
257. 220 FX 0 FY -3.62 FZ -1.2
258. 146 FX 0.01 FY 2.74 FZ -0.01
259. 222 FX 0 FY 2.75 FZ -0.01
260. ********************************************************************888
261. ********************************************************************************
262. LOAD 100 1.4D
263. *SOIL+SURCHARGE=2.2X1.8*1.4+1=6.5T/SQ.M)
264. SELFWEIGHT Y -1
265. ELEMENT LOAD
266. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
267. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
268. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
269. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -6.5
270. JOINT LOAD
271. 145 FX 0 FY -1.26 FZ 0
272. 220 FX 0 FY -1.02 FZ 0
273. 146 FX 0 FY -1.27 FZ 0
274. 222 FX 0 FY -1.01 FZ 0
275. ********************************************************************************
276. LOAD 101 1.4D+1.7(LL)
277. SELFWEIGHT Y -1
278. ELEMENT LOAD
279. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
280. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
281. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
282. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -6.5
283. JOINT LOAD
284. 145 FX 1.26 FY 11.57 FZ 2.94
285. 220 FX 0 FY 17.07 FZ 6.67
286. 146 FX -2.41 FY -19.84 FZ -0.02
287. 222 FX 0 FY -17.85 FZ -0.02
288. ********************************************************************************
289. LOAD 102 1.2D+1.0(LL)+1.6WL(+X)
290. SELFWEIGHT Y -1
291. ELEMENT LOAD
292. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
293. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
294. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
295. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5.8
296. JOINT LOAD
297. 145 FX 1.63 FY 7.88 FZ 1.73
298. 220 FX 0.02 FY 8.35 FZ 3.92
299. 146 FX -0.53 FY -10.64 FZ -0.02
300. 222 FX 0.03 FY -12.15 FZ -0.01
301. ********************************************************************************
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302. LOAD 103 1.2D+1.0(LL)+1.6WL(-X)
303. SELFWEIGHT Y -1
304. ELEMENT LOAD
305. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
306. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
307. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
308. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5.8
309. JOINT LOAD
310. 145 FX -0.15 FY 5.05 FZ 1.73
311. 220 FX -0.02 FY 11.18 FZ 3.92
312. 146 FX -2.3 FY -13.39 FZ -0.01
313. 222 FX -0.02 FY -9.39 FZ -0.02
314. ******************************************************************************
315. LOAD 104 1.2D+1.0(LL)+1.6WL(+Z)
316. SELFWEIGHT Y -1
317. ELEMENT LOAD
318. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
319. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
320. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
321. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5.8
322. JOINT LOAD
323. 145 FX 0.76 FY 11.72 FZ 3.67
324. 220 FX 0 FY 14.86 FZ 5.84
325. 146 FX -1.43 FY -17.27 FZ 0
326. 222 FX 0 FY -15.87 FZ 0
327. ******************************************************************************
328. LOAD 105 1.2D+1.0(LL)+1.6WL(-Z)
329. SELFWEIGHT Y -1
330. ELEMENT LOAD
331. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
332. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
333. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
334. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -5.8
335. JOINT LOAD
336. 145 FX 0.73 FY 1.21 FZ -0.21
337. 220 FX 0 FY 4.67 FZ 2
338. 146 FX -1.4 FY -6.76 FZ -0.03
339. 222 FX 0 FY -5.67 FZ -0.03
340. ******************************************************************************
341. LOAD 106 0.9*D+1.6WL(+X)
342. SELFWEIGHT Y -1
343. ELEMENT LOAD
344. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
345. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
346. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
347. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -4.6
348. JOINT LOAD
349. 145 FX 0.89 FY 0.6 FZ 0
350. 220 FX 0.02 FY -2.07 FZ 0
351. 146 FX 0.89 FY 0.56 FZ 0
352. 222 FX 0.03 FY -2.03 FZ 0
353. ******************************************************************************
354. LOAD 107 0.9*D+1.6WL(-X)
355. SELFWEIGHT Y -1
356. ELEMENT LOAD
357. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
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358. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
359. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
360. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -4.6
361. JOINT LOAD
362. 145 FX -0.89 FY -2.23 FZ 0
363. 220 FX -0.02 FY 0.76 FZ 0
364. 146 FX -0.89 FY -2.19 FZ 0
365. 222 FX -0.03 FY 0.73 FZ 0
366. ******************************************************************************
367. LOAD 108 0.9*D+1.6WL(+Z)
368. SELFWEIGHT Y -1
369. ELEMENT LOAD
370. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
371. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
372. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
373. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -4.6
374. JOINT LOAD
375. 145 FX 0.02 FY 4.44 FZ 1.94
376. 220 FX 0 FY 4.44 FZ 1.92
377. 146 FX -0.02 FY -6.07 FZ 0.02
378. 222 FX 0 FY -5.75 FZ 0.02
379. ******************************************************************************
380. LOAD 109 0.9*D+1.6WL(-Z)
381. SELFWEIGHT Y -1
382. ELEMENT LOAD
383. 25 TO 44 65 TO 76 81 82 85 86 89 90 93 94 97 98 121 122 125 126 129 130 135 -
384. 136 TO 140 143 TO 148 335 TO 339 345 TO 349 375 376 378 379 382 385 TO 395 -
385. 405 TO 412 422 423 458 TO 462 464 TO 470 523 TO 528 530 531 545 555 TO 560 -
386. 562 563 577 587 TO 591 593 TO 600 602 TO 616 PR GY -4.6
387. JOINT LOAD
388. 145 FX -0.02 FY -6.07 FZ -1.94
389. 220 FX 0 FY -5.75 FZ -1.92
390. 146 FX 0.02 FY 4.44 FZ -0.02
391. 222 FX 0 FY 4.45 FZ -0.02
392. ******************************************************************************
393. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**184**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**154**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**180**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 141/ 21/ 69 DOF
TOTAL PRIMARY LOAD CASES = 19, TOTAL DEGREES OF FREEDOM = 564
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 39 DOUBLE KILO-WORDS

REQRD/AVAIL. DISK SPACE = 13.4/ 164307.4 MB

394. LOAD LIST 100 TO 109
395. START CONCRETE DESIGN

```
396. CODE ACI 1999
397. TRACK 0 MEMB 133 134
398. MAXMAIN 25 MEMB 133 134
399. CLB 0.075 MEMB 133 134
400. CLS 0.075 MEMB 133 134
401. DESIGN COLUMN 133 134
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

C01

ATTACHMENT 4 STAAD Input & Output File of Foundation Mkd. F3

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    AUG  6, 2020                *
*           Time=    19:10:42                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F3.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 0; 14 0 -4 -0.25; 16 0 -4 1.2; 20 1 -4 1.2; 21 0.75 -4 1.2
     9. 22 0.75 -4 0.95; 23 1 -4 0.95; 24 0.5 -4 1.2; 25 0.5 -4 0.95; 26 0.25 -4 1.2
    10. 27 0.25 -4 0.95; 28 0 -4 0.95; 29 0.75 -4 0.5; 30 1 -4 0.5; 31 0.5 -4 0.5
    11. 32 0.25 -4 0.5; 33 0 -4 0.5; 34 0.75 -4 0.25; 35 1 -4 0.25; 36 0.5 -4 0.25
    12. 37 0.25 -4 0.25; 38 0 -4 0.25; 39 0.75 -4 0; 40 1 -4 0; 41 0.5 -4 0
    13. 42 0.25 -4 0; 43 0.75 -4 -0.25; 44 1 -4 -0.25; 45 0.5 -4 -0.25
    14. 46 0.25 -4 -0.25; 72 0.75 -4 -0.5; 73 1 -4 -0.5; 74 0.5 -4 -0.5
    15. 75 0.25 -4 -0.5; 76 0 -4 -0.5; 77 0.75 -4 -0.75; 78 1 -4 -0.75
    16. 79 0.5 -4 -0.75; 80 0.25 -4 -0.75; 81 0 -4 -0.75; 82 0.75 -4 -1; 83 1 -4 -1.
    17. 84 0.5 -4 -1; 85 0.25 -4 -1; 86 0 -4 -1; 90 -0.25 -4 1.2; 91 -0.25 -4 0.95
    18. 92 -0.5 -4 1.2; 93 -0.5 -4 0.95; 97 -0.25 -4 0.5; 98 -0.5 -4 0.5
    19. 101 -0.25 -4 0.25; 102 -0.5 -4 0.25; 105 -0.25 -4 0; 106 -0.5 -4 0
    20. 109 -0.25 -4 -0.25; 110 -0.5 -4 -0.25; 133 -0.25 -4 -0.5; 134 -0.5 -4 -0.5
    21. 137 -0.25 -4 -0.75; 138 -0.5 -4 -0.75; 141 -0.25 -4 -1; 142 -0.5 -4 -1.
    22. 145 0 -1 -0.75; 146 0 -1 0.95; 147 0.75 -4 -1.25; 148 1 -4 -1.25
    23. 149 0.5 -4 -1.25; 150 0.25 -4 -1.25; 151 0 -4 -1.25; 152 -0.25 -4 -1.25
    24. 153 -0.5 -4 -1.25; 156 1 -4 1.45; 157 0.75 -4 1.45; 158 0.5 -4 1.45
    25. 159 0.25 -4 1.45; 160 0 -4 1.45; 161 -0.25 -4 1.45; 162 -0.5 -4 1.45
    26. 181 1.5 -4 1.2; 182 1.5 -4 0.95; 183 1.5 -4 0.5; 184 1.5 -4 0.25; 185 1.5 -4 0
    27. 186 1.5 -4 -0.25; 192 1.5 -4 -0.5; 193 1.5 -4 -0.75; 194 1.5 -4 -1.
    28. 195 1.5 -4 -1.25; 196 1.5 -4 1.45; 220 1.7 -1 -0.75; 222 1.7 -1 0.95
    29. 361 1.7 -4 1.2; 362 1.7 -4 0.95; 363 1.7 -4 0.5; 364 1.7 -4 0.25; 365 1.7 -4 0
    30. 366 1.7 -4 -0.25; 372 1.7 -4 -0.5; 373 1.7 -4 -0.75; 374 1.7 -4 -1.
    31. 375 1.7 -4 -1.25; 376 1.7 -4 1.45; 419 1.95 -4 1.2; 420 1.95 -4 0.95
    32. 421 1.95 -4 0.5; 422 1.95 -4 0.25; 423 1.95 -4 0; 424 1.95 -4 -0.25
    33. 430 1.95 -4 -0.5; 431 1.95 -4 -0.75; 432 1.95 -4 -1; 433 1.95 -4 -1.25
    34. 434 1.95 -4 1.45; 435 0.75 -4 -1.5; 436 1 -4 -1.5; 437 0.5 -4 -1.5
    35. 438 0.25 -4 -1.5; 439 0 -4 -1.5; 440 -0.25 -4 -1.5; 441 -0.5 -4 -1.5
    36. 442 1.5 -4 -1.5; 451 1.7 -4 -1.5; 452 1.95 -4 -1.5; 453 1 -4 1.7
    37. 454 0.75 -4 1.7; 455 0.5 -4 1.7; 456 0.25 -4 1.7; 457 0 -4 1.7
    38. 458 -0.25 -4 1.7; 459 -0.5 -4 1.7; 460 1.5 -4 1.7; 469 1.7 -4 1.7
```

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39. 470 1.95 -4 1.7; 507 1.25 -4 1.2; 508 1.25 -4 0.95; 509 1.25 -4 0.5
40. 510 1.25 -4 0.25; 511 1.25 -4 0; 512 1.25 -4 -0.25; 513 1.25 -4 -0.5
41. 514 1.25 -4 -0.75; 515 1.25 -4 -1; 516 1.25 -4 -1.25; 517 1.25 -4 1.45
42. 518 1.25 -4 -1.5; 519 1.25 -4 1.7; 631 2.2 -4 1.2; 632 2.2 -4 0.95
43. 633 2.2 -4 0.5; 634 2.2 -4 0.25; 635 2.2 -4 0; 636 2.2 -4 -0.25
44. 637 2.2 -4 -0.5; 638 2.2 -4 -0.75; 639 2.2 -4 -1; 640 2.2 -4 -1.25
45. 641 2.2 -4 1.45; 642 2.2 -4 -1.5; 643 2.2 -4 1.7; 646 1 -4 0.75
46. 647 0.75 -4 0.75; 648 0.5 -4 0.75; 649 0.25 -4 0.75; 650 0 -4 0.75
47. 651 -0.25 -4 0.75; 652 -0.5 -4 0.75; 653 1.7 -4 0.75; 654 1.5 -4 0.75
48. 655 1.95 -4 0.75; 656 1.25 -4 0.75; 657 2.2 -4 0.75; 658 0.75 -4 -1.575
49. 659 1 -4 -1.575; 660 0.5 -4 -1.575; 661 0.25 -4 -1.575; 662 0 -4 -1.575
50. 663 -0.25 -4 -1.575; 664 -0.5 -4 -1.575; 665 1.5 -4 -1.575; 666 1.7 -4 -1.575
51. 667 1.95 -4 -1.575; 668 1.25 -4 -1.575; 669 2.2 -4 -1.575; 670 1 -4 1.775
52. 671 0.75 -4 1.775; 672 0.5 -4 1.775; 673 0.25 -4 1.775; 674 0 -4 1.775
53. 675 -0.25 -4 1.775; 676 -0.5 -4 1.775; 677 1.7 -4 1.775; 678 1.5 -4 1.775
54. 679 1.95 -4 1.775; 680 1.25 -4 1.775; 681 2.2 -4 1.775; 682 -0.75 -4 1.2
55. 683 -0.75 -4 0.95; 684 -0.75 -4 0.5; 685 -0.75 -4 0.25; 686 -0.75 -4 0
56. 687 -0.75 -4 -0.25; 688 -0.75 -4 -0.5; 689 -0.75 -4 -0.75; 690 -0.75 -4 -1.
57. 691 -0.825 -4 1.2; 692 -0.825 -4 0.95; 695 -0.825 -4 0.5; 696 -0.825 -4 0.25
58. 699 -0.825 -4 0; 701 -0.825 -4 -0.25; 703 -0.825 -4 -0.5; 705 -0.825 -4 -0.75
59. 707 -0.825 -4 -1; 709 -0.75 -4 -1.25; 710 -0.825 -4 -1.25; 712 -0.75 -4 1.45
60. 713 -0.825 -4 1.45; 715 -0.75 -4 -1.5; 716 -0.825 -4 -1.5; 718 -0.75 -4 1.7
61. 719 -0.825 -4 1.7; 721 -0.75 -4 0.75; 722 -0.825 -4 0.75; 724 -0.75 -4 -1.575
62. 725 -0.825 -4 -1.575; 727 -0.75 -4 1.775; 728 -0.825 -4 1.775; 729 2.45 -4 1.2
63. 730 2.45 -4 0.95; 731 2.45 -4 0.25; 732 2.45 -4 0; 733 2.45 -4 -0.25
64. 734 2.45 -4 0.5; 735 2.45 -4 -0.5; 736 2.45 -4 -0.75; 737 2.45 -4 -1.
65. 738 2.45 -4 -1.25; 739 2.45 -4 1.45; 740 2.45 -4 -1.5; 741 2.45 -4 1.7
66. 742 2.525 -4 1.2; 743 2.525 -4 0.95; 744 2.525 -4 0.5; 745 2.525 -4 0.25
67. 746 2.525 -4 0; 747 2.525 -4 -0.25; 748 2.525 -4 -0.5; 749 2.525 -4 -0.75
68. 750 2.525 -4 -1; 751 2.525 -4 -1.25; 752 2.525 -4 1.45; 753 2.525 -4 -1.5
69. 754 2.525 -4 1.7; 755 2.45 -4 0.75; 756 2.525 -4 0.75; 757 2.45 -4 -1.575
70. 758 2.525 -4 -1.575; 759 2.45 -4 1.775; 760 2.525 -4 1.775
71. MEMBER INCIDENCES
72. 133 81 145; 134 28 146; 201 373 220; 202 362 222
73. ELEMENT INCIDENCES SHELL
74. 25 20 21 22 23; 26 21 24 25 22; 27 24 26 27 25; 28 26 16 28 27; 33 30 29 34 35
75. 34 29 31 36 34; 35 31 32 37 36; 36 32 33 38 37; 37 35 34 39 40; 38 34 36 41 39
76. 39 36 37 42 41; 40 37 38 13 42; 41 40 39 43 44; 42 39 41 45 43; 43 41 42 46 45
77. 44 42 13 14 46; 65 44 43 72 73; 66 43 45 74 72; 67 45 46 75 74; 68 46 14 76 75
78. 69 73 72 77 78; 70 72 74 79 77; 71 74 75 80 79; 72 75 76 81 80; 73 78 77 82 83
79. 74 77 79 84 82; 75 79 80 85 84; 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91
80. 89 33 97 101 38; 90 97 98 102 101; 93 38 101 105 13; 94 101 102 106 105
81. 97 13 105 109 14; 98 105 106 110 109; 121 14 109 133 76; 122 109 110 134 133
82. 125 76 133 137 81; 126 133 134 138 137; 129 81 137 141 86; 130 137 138 142 141
83. 135 83 82 147 148; 136 82 84 149 147; 137 84 85 150 149; 138 85 86 151 150
84. 139 86 141 152 151; 140 141 142 153 152; 143 156 157 21 20; 144 157 158 24 21
85. 145 158 159 26 24; 146 159 160 16 26; 147 160 161 90 16; 148 161 162 92 90
86. 335 361 181 182 362; 337 363 183 184 364; 338 364 184 185 365
87. 339 365 185 186 366; 345 366 186 192 372; 346 372 192 193 373
88. 347 373 193 194 374; 348 374 194 195 375; 349 376 196 181 361
89. 375 419 361 362 420; 378 422 364 365 423; 379 423 365 366 424
90. 382 421 363 364 422; 385 424 366 372 430; 386 430 372 373 431
91. 387 431 373 374 432; 388 432 374 375 433; 389 434 376 361 419
92. 390 148 147 435 436; 391 147 149 437 435; 392 149 150 438 437
93. 393 150 151 439 438; 394 151 152 440 439; 395 152 153 441 440
94. 405 375 195 442 451; 406 433 375 451 452; 407 453 454 157 156
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 95. 408 454 455 158 157; 409 455 456 159 158; 410 456 457 160 159
 96. 411 457 458 161 160; 412 458 459 162 161; 422 469 460 196 376
 97. 423 470 469 376 434; 458 507 20 23 508; 460 509 30 35 510; 461 510 35 40 511
 98. 462 511 40 44 512; 464 512 44 73 513; 465 513 73 78 514; 466 514 78 83 515
 99. 467 515 83 148 516; 468 517 156 20 507; 469 516 148 436 518
100. 470 519 453 156 517; 587 181 507 508 182; 589 183 509 510 184
101. 590 184 510 511 185; 591 185 511 512 186; 593 186 512 513 192
102. 594 192 513 514 193; 595 193 514 515 194; 596 194 515 516 195
103. 597 196 517 507 181; 598 195 516 518 442; 599 460 519 517 196
104. 603 631 419 420 632; 605 633 421 422 634; 606 634 422 423 635
105. 607 635 423 424 636; 608 636 424 430 637; 609 637 430 431 638
106. 610 638 431 432 639; 611 639 432 433 640; 612 641 434 419 631
107. 613 640 433 452 642; 614 643 470 434 641; 617 646 647 29 30; 618 647 648 31 29
108. 619 648 649 32 31; 620 649 650 33 32; 621 650 651 97 33; 622 651 652 98 97
109. 623 653 654 183 363; 624 655 653 363 421; 625 656 646 30 509
110. 626 654 656 509 183; 627 657 655 421 633; 628 652 651 91 93; 629 651 650 28 91
111. 630 650 649 27 28; 631 649 648 25 27; 632 648 647 22 25; 633 647 646 23 22
112. 634 646 656 508 23; 635 656 654 182 508; 636 654 653 362 182
113. 637 653 655 420 362; 638 655 657 632 420; 639 436 435 658 659
114. 640 435 437 660 658; 641 437 438 661 660; 642 438 439 662 661
115. 643 439 440 663 662; 644 440 441 664 663; 645 451 442 665 666
116. 646 452 451 666 667; 647 518 436 659 668; 648 442 518 668 665
117. 649 642 452 667 669; 650 670 671 454 453; 651 671 672 455 454
118. 652 672 673 456 455; 653 673 674 457 456; 654 674 675 458 457
119. 655 675 676 459 458; 656 677 678 460 469; 657 679 677 469 470
120. 658 680 670 453 519; 659 678 680 519 460; 660 681 679 470 643
121. 661 92 682 683 93; 662 98 684 685 102; 663 102 685 686 106
122. 664 106 686 687 110; 665 110 687 688 134; 666 134 688 689 138
123. 667 138 689 690 142; 668 682 691 692 683; 670 684 695 696 685
124. 672 685 696 699 686; 674 686 699 701 687; 676 687 701 703 688
125. 678 688 703 705 689; 680 689 705 707 690; 682 142 690 709 153
126. 683 690 707 710 709; 685 162 712 682 92; 686 712 713 691 682
127. 688 153 709 715 441; 689 709 710 716 715; 691 459 718 712 162
128. 692 718 719 713 712; 694 652 721 684 98; 695 721 722 695 684
129. 698 722 721 683 692; 699 721 652 93 683; 700 441 715 724 664
130. 701 715 716 725 724; 703 676 727 718 459; 704 727 728 719 718
131. 705 729 631 632 730; 706 731 634 635 732; 707 732 635 636 733
132. 708 734 633 634 731; 709 733 636 637 735; 710 735 637 638 736
133. 711 736 638 639 737; 712 737 639 640 738; 713 739 641 631 729
134. 714 738 640 642 740; 715 741 643 641 739; 716 742 729 730 743
135. 717 744 734 731 745; 718 745 731 732 746; 719 746 732 733 747
136. 720 747 733 735 748; 721 748 735 736 749; 722 749 736 737 750
137. 723 750 737 738 751; 724 752 739 729 742; 725 751 738 740 753
138. 726 754 741 739 752; 727 755 657 633 734; 728 756 755 734 744
139. 729 657 755 730 632; 730 755 756 743 730; 731 740 642 669 757
140. 732 753 740 757 758; 733 759 681 643 741; 734 760 759 741 754
141. START GROUP DEFINITION
142. ELEMENT
143. _MOMENT 25 TO 27 33 TO 44 65 TO 71 73 TO 75 82 89 90 93 94 97 98 121 122 126 -
144. 130 135 TO 140 143 TO 148 337 TO 339 345 348 349 378 379 382 385 388 TO 395 -
145. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
146. 605 TO 614 617 TO 628 631 TO 635 638 TO 668 670 672 674 676 678 680 682 683 -
147. 685 686 688 689 691 692 694 695 698 TO 701 703 TO 734
148. _SHEAR 25 33 37 TO 41 65 69 73 93 94 135 143 338 378 390 407 461 590 606 617 -
149. 633 639 TO 660 663 668 670 672 674 676 678 680 683 686 689 692 695 698 700 -
150. 701 703 704 706 716 TO 726 728 730 TO 734
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151. END GROUP DEFINITION
152. ELEMENT PROPERTY
153. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
154. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
155. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
156. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
157. 692 694 695 698 TO 701 703 TO 734 THICKNESS 0.6
158. SUPPORTS
159. 13 14 16 20 TO 46 72 TO 86 90 TO 93 97 98 101 102 105 106 109 110 133 134 -
160. 137 138 141 142 147 TO 153 156 TO 162 181 TO 186 192 TO 196 361 TO 366 372 -
161. 373 TO 376 419 TO 424 430 TO 442 451 TO 460 469 470 507 TO 519 631 TO 643 -
162. 646 TO 692 695 696 699 701 703 705 707 709 710 712 713 715 716 718 719 721 -
163. 722 724 725 727 TO 760 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
```

JOINT	INFLUENCE AREA
13	0.062500
14	0.062500
16	0.062500
20	0.062500
21	0.062500
22	0.056250
23	0.056250
24	0.062500
25	0.056250
26	0.062500
27	0.056250
28	0.056250
29	0.062500
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.062500
36	0.062500
37	0.062500
38	0.062500
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.062500
45	0.062500
46	0.062500
72	0.062500
73	0.062500
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062500
80	0.062500

81	0.062500
82	0.062500
83	0.062500
84	0.062500
85	0.062500
86	0.062500
90	0.062500
91	0.056250
92	0.062500
93	0.056250
97	0.062500
98	0.062500
101	0.062500
102	0.062500
105	0.062500
106	0.062500
109	0.062500
110	0.062500
133	0.062500
134	0.062500
137	0.062500
138	0.062500
141	0.062500
142	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.062500
151	0.062500
152	0.062500
153	0.062500
156	0.062500
157	0.062500
158	0.062500
159	0.062500
160	0.062500
161	0.062500
162	0.062500
181	0.056250
182	0.050625
183	0.056250
184	0.056250
185	0.056250
186	0.056250
192	0.056250
193	0.056250
194	0.056250
195	0.056250
196	0.056250
361	0.056250
362	0.050625
363	0.056250
364	0.056250
365	0.056250
366	0.056250
372	0.056250

373	0.056250
374	0.056250
375	0.056250
376	0.056250
419	0.062500
420	0.056250
421	0.062500
422	0.062500
423	0.062500
424	0.062500
430	0.062500
431	0.062500
432	0.062500
433	0.062500
434	0.062500
435	0.040625
436	0.040625
437	0.040625
438	0.040625
439	0.040625
440	0.040625
441	0.040625
442	0.036562
451	0.036562
452	0.040625
453	0.040625
454	0.040625
455	0.040625
456	0.040625
457	0.040625
458	0.040625
459	0.040625
460	0.036563
469	0.036563
470	0.040625
507	0.062500
508	0.056250
509	0.062500
510	0.062500
511	0.062500
512	0.062500
513	0.062500
514	0.062500
515	0.062500
516	0.062500
517	0.062500
518	0.040625
519	0.040625
631	0.062500
632	0.056250
633	0.062500
634	0.062500
635	0.062500
636	0.062500
637	0.062500
638	0.062500

639	0.062500
640	0.062500
641	0.062500
642	0.040625
643	0.040625
646	0.056250
647	0.056250
648	0.056250
649	0.056250
650	0.056250
651	0.056250
652	0.056250
653	0.050625
654	0.050625
655	0.056250
656	0.056250
657	0.056250
658	0.009375
659	0.009375
660	0.009375
661	0.009375
662	0.009375
663	0.009375
664	0.009375
665	0.008437
666	0.008437
667	0.009375
668	0.009375
669	0.009375
670	0.009375
671	0.009375
672	0.009375
673	0.009375
674	0.009375
675	0.009375
676	0.009375
677	0.008438
678	0.008438
679	0.009375
680	0.009375
681	0.009375
682	0.040625
683	0.036563
684	0.040625
685	0.040625
686	0.040625
687	0.040625
688	0.040625
689	0.040625
690	0.040625
691	0.009375
692	0.008438
695	0.009375
696	0.009375
699	0.009375
701	0.009375

703	0.009375
705	0.009375
707	0.009375
709	0.040625
710	0.009375
712	0.040625
713	0.009375
715	0.026406
716	0.006094
718	0.026406
719	0.006094
721	0.036562
722	0.008438
724	0.006094
725	0.001406
727	0.006094
728	0.001406
729	0.040625
730	0.036563
731	0.040625
732	0.040625
733	0.040625
734	0.040625
735	0.040625
736	0.040625
737	0.040625
738	0.040625
739	0.040625
740	0.026406
741	0.026406
742	0.009375
743	0.008438
744	0.009375
745	0.009375
746	0.009375
747	0.009375
748	0.009375
749	0.009375
750	0.009375
751	0.009375
752	0.009375
753	0.006094
754	0.006094
755	0.036563
756	0.008438
757	0.006094
758	0.001406
759	0.006094
760	0.001406
TOTAL	11.222477

```
164. DEFINE MATERIAL START
165. ISOTROPIC STEEL
166. E 2.09042E+007
167. POISSON 0.3
168. DENSITY 7.83341
169. ALPHA 1.2E-005
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170. DAMP 0.03
171. TYPE STEEL
172. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
173. ISOTROPIC CONCRETE
174. E 2.21467E+006
175. POISSON 0.17
176. DENSITY 2.40262
177. ALPHA 1E-005
178. DAMP 0.05
179. TYPE CONCRETE
180. STRENGTH FCU 2812.28
181. END DEFINE MATERIAL
182. MEMBER PROPERTY AMERICAN
183. 133 134 201 202 PRIS YD 0.6 ZD 0.6
184. CONSTANTS
185. MATERIAL CONCRETE ALL
186. ******************************************************************************
187. LOAD 1 DL+LL
188. SELFWEIGHT Y -1
189. ELEMENT LOAD
190. *SOIL+SURCHARGE=2.2X1.8+1=5 T/SQ.M)
191. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
192. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
193. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
194. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
195. 692 694 695 698 TO 701 703 TO 734 PR GY -5
196. JOINT LOAD
197. 145 FX 3.11 FY 18.74 FZ 0
198. 220 FX 2.4 FY -21 FZ -6.87
199. 146 FX -0.18 FY 6.49 FZ 0.01
200. 222 FX 0.08 FY -37.23 FZ -7.32
201. ******************************************************************************8
202. LOAD 2 DL+0.75*LL+0.75*WL1
203. SELFWEIGHT Y -1
204. ELEMENT LOAD
205. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
206. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
207. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
208. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
209. 692 694 695 698 TO 701 703 TO 734 PR GY -5
210. JOINT LOAD
211. 145 FX 3.17 FY 14.93 FZ 0
212. 220 FX 2.69 FY -14.98 FZ -5.15
213. 146 FX -0.12 FY 2.96 FZ 0.01
214. 222 FX 0.08 FY -29.97 FZ -5.5
215. ******************************************************************************8
216. LOAD 3 DL+0.75*LL+0.75*WL2
217. SELFWEIGHT Y -1
218. ELEMENT LOAD
219. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
220. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
221. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
222. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
223. 692 694 695 698 TO 701 703 TO 734 PR GY -5
224. JOINT LOAD
225. 145 FX 1.46 FY 11.96 FZ 0
```

```
226. 220 FX 0.92 FY -18.01 FZ -5.15
227. 146 FX -0.14 FY 5.92 FZ 0.01
228. 222 FX 0.05 FY -26.94 FZ -5.49
229. ******************************************************************************
230. LOAD 4 DL+0.75*LL+0.75*WL3
231. SELFWEIGHT Y -1
232. ELEMENT LOAD
233. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
234. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
235. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
236. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
237. 692 694 695 698 TO 701 703 TO 734 PR GY -5
238. JOINT LOAD
239. 145 FX 2.28 FY 11.33 FZ 0.02
240. 220 FX 1.85 FY -14.38 FZ -4.22
241. 146 FX -0.12 FY 2.16 FZ 0.03
242. 222 FX 0.05 FY -26.18 FZ -4.56
243. ******************************************************************************
244. LOAD 5 DL+0.75*LL+0.75*WL4
245. SELFWEIGHT Y -1
246. ELEMENT LOAD
247. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
248. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
249. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
250. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
251. 692 694 695 698 TO 701 703 TO 734 PR GY -5
252. JOINT LOAD
253. 145 FX 2.35 FY 15.56 FZ -0.02
254. 220 FX 1.76 FY -18.62 FZ -6.08
255. 146 FX -0.14 FY 6.73 FZ -0.01
256. 222 FX 0.08 FY -30.74 FZ -6.43
257. ******************************************************************************
258. LOAD 6 0.6*DL+WL1
259. SELFWEIGHT Y -1
260. ELEMENT LOAD
261. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
262. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
263. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
264. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
265. 692 694 695 698 TO 701 703 TO 734 PR GY -3.4
266. JOINT LOAD
267. 145 FX 1.09 FY 0.5 FZ 0
268. 220 FX 0.03 FY -3 FZ 0
269. 146 FX 1.19 FY 0.22 FZ 0.02
270. 222 FX 0.03 FY -3.3 FZ -0.02
271. ******************************************************************************
272. LOAD 7 0.6*DL+WL2
273. SELFWEIGHT Y -1
274. ELEMENT LOAD
275. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
276. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
277. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
278. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
279. 692 694 695 698 TO 701 703 TO 734 PR GY -3.4
280. JOINT LOAD
281. 145 FX -1.2 FY -3.45 FZ 0
```

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282. 220 FX -1.18 FY -3.82 FZ 0.01
283. 146 FX 0 FY 0.95 FZ 0
284. 222 FX 0 FY 0.74 FZ 0
285. ******************************************************************************
286. LOAD 8 0.6*DL+WL3
287. SELFWEIGHT Y -1
288. ELEMENT LOAD
289. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
290. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
291. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
292. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
293. 692 694 695 698 TO 701 703 TO 734 PR GY -3.4
294. JOINT LOAD
295. 145 FX -0.1 FY -4.3 FZ 0.03
296. 220 FX 0.06 FY 1.03 FZ 1.25
297. 146 FX 0.03 FY -4.07 FZ 0.03
298. 222 FX 0 FY 1.76 FZ 1.23
299. ******************************************************************************
300. LOAD 9 0.6*DL+WL4
301. SELFWEIGHT Y -1
302. ELEMENT LOAD
303. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
304. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
305. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
306. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
307. 692 694 695 698 TO 701 703 TO 734 PR GY -3.4
308. JOINT LOAD
309. 145 FX 0 FY 1.35 FZ -0.02
310. 220 FX -0.06 FY -4.63 FZ -1.22
311. 146 FX -0.01 FY 2.02 FZ -0.02
312. 222 FX 0.03 FY -4.32 FZ -1.25
313. ******************************************************************888
314. ******************************************************************************
315. LOAD 100 1.4D
316. *SOIL+SURCHARGE=2.2X1.8*1.4+1=6.5T/SQ.M)
317. SELFWEIGHT Y -1
318. ELEMENT LOAD
319. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
320. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 390 TO 395 405 407 TO 412 -
321. 422 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 617 TO 623 625 626 -
322. 639 TO 645 647 648 650 TO 656 658 659 661 TO 668 670 672 674 676 678 680 -
323. 682 683 685 686 688 689 691 692 694 695 700 701 703 704 PR GY -6.5
324. JOINT LOAD
325. 145 FX -0.12 FY -3.44 FZ 0
326. 220 FX 0.01 FY -4.2 FZ 0.04
327. 146 FX 0.03 FY -2.39 FZ 0
328. 222 FX 0.03 FY -2.99 FZ -0.03
329. ******************************************************************************
330. LOAD 101 1.4D+1.7(LL)
331. SELFWEIGHT Y -1
332. ELEMENT LOAD
333. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
334. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
335. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
336. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
337. 692 694 695 698 TO 701 703 TO 734 PR GY -6.5
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338. JOINT LOAD
339. 145 FX 5.32 FY 32.6 FZ 0
340. 220 FX 4.09 FY -34.8 FZ -11.69
341. 146 FX -0.31 FY 11.55 FZ 0.01
342. 222 FX 0.13 FY -62.66 FZ -12.44
343. ******************************************************************************
344. LOAD 102 1.2D+1.0(LL)+1.6WL(+X)
345. SELFWEIGHT Y -1
346. ELEMENT LOAD
347. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
348. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
349. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
350. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
351. 692 694 695 698 TO 701 703 TO 734 PR GY -5.8
352. JOINT LOAD
353. 145 FX 4.93 FY 21.42 FZ 0
354. 220 FX 4.3 FY -18.37 FZ -6.86
355. 146 FX -0.15 FY 2.99 FZ 0.01
356. 222 FX 0.11 FY -40.89 FZ -7.33
357. ******************************************************************************
358. LOAD 103 1.2D+1.0(LL)+1.6WL(-X)
359. SELFWEIGHT Y -1
360. ELEMENT LOAD
361. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
362. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
363. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
364. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
365. 692 694 695 698 TO 701 703 TO 734 PR GY -5.8
366. JOINT LOAD
367. 145 FX 1.26 FY 15.09 FZ 0
368. 220 FX 0.51 FY -24.82 FZ -6.87
369. 146 FX -0.2 FY 9.31 FZ 0.01
370. 222 FX 0.06 FY -34.43 FZ -7.31
371. ******************************************************************************
372. LOAD 104 1.2D+1.0(LL)+1.6WL(+Z)
373. SELFWEIGHT Y -1
374. ELEMENT LOAD
375. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
376. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
377. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
378. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
379. 692 694 695 698 TO 701 703 TO 734 PR GY -5.8
380. JOINT LOAD
381. 145 FX 3.02 FY 13.73 FZ 0.04
382. 220 FX 2.5 FY -17.07 FZ -4.89
383. 146 FX -0.15 FY 1.28 FZ 0.05
384. 222 FX 0.06 FY -32.8 FZ -5.33
385. ******************************************************************************
386. LOAD 105 1.2D+1.0(LL)+1.6WL(-Z)
387. SELFWEIGHT Y -1
388. ELEMENT LOAD
389. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
390. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
391. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
392. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
393. 692 694 695 698 TO 701 703 TO 734 PR GY -5.8
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394. JOINT LOAD
395. 145 FX 3.18 FY 22.77 FZ -0.04
396. 220 FX 2.31 FY -26.12 FZ -8.85
397. 146 FX -0.21 FY 11.03 FZ -0.03
398. 222 FX 0.11 FY -42.53 FZ -9.31
399. ******************************************************************************
400. LOAD 106 0.9*D+1.6WL(+X)
401. SELFWEIGHT Y -1
402. ELEMENT LOAD
403. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
404. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
405. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
406. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
407. 692 694 695 698 TO 701 703 TO 734 PR GY -4.6
408. JOINT LOAD
409. 145 FX 1.75 FY 0.96 FZ 0
410. 220 FX 1.9 FY 0.53 FZ 0.03
411. 146 FX 0.04 FY -4.7 FZ 0
412. 222 FX 0.05 FY -5.15 FZ -0.03
413. ******************************************************************************
414. LOAD 107 0.9*D+1.6WL(-X)
415. SELFWEIGHT Y -1
416. ELEMENT LOAD
417. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
418. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
419. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
420. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
421. 692 694 695 698 TO 701 703 TO 734 PR GY -4.6
422. JOINT LOAD
423. 145 FX -1.91 FY -5.38 FZ 0
424. 220 FX -1.89 FY -5.93 FZ 0.02
425. 146 FX -0.01 FY 1.63 FZ 0
426. 222 FX 0 FY 1.31 FZ 0
427. ******************************************************************************
428. LOAD 108 0.9*D+1.6WL(+Z)
429. SELFWEIGHT Y -1
430. ELEMENT LOAD
431. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
432. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
433. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
434. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
435. 692 694 695 698 TO 701 703 TO 734 PR GY -4.6
436. JOINT LOAD
437. 145 FX -0.16 FY -6.73 FZ 0.04
438. 220 FX 0.1 FY 1.83 FZ 2
439. 146 FX 0.04 FY -6.41 FZ 0.04
440. 222 FX -0.01 FY 2.95 FZ 1.97
441. ******************************************************************************
442. LOAD 109 0.9*D+1.6WL(-Z)
443. SELFWEIGHT Y -1
444. ELEMENT LOAD
445. 25 TO 28 33 TO 44 65 TO 76 81 82 89 90 93 94 97 98 121 122 125 126 129 130 -
446. 135 TO 140 143 TO 148 335 337 TO 339 345 TO 349 375 378 379 382 385 TO 395 -
447. 405 TO 412 422 423 458 460 TO 462 464 TO 470 587 589 TO 591 593 TO 599 603 -
448. 605 TO 614 617 TO 668 670 672 674 676 678 680 682 683 685 686 688 689 691 -
449. 692 694 695 698 TO 701 703 TO 734 PR GY -4.6
```

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450. JOINT LOAD
451. 145 FX 0 FY 2.31 FZ -0.04
452. 220 FX -0.09 FY -7.23 FZ -1.96
453. 146 FX -0.01 FY 3.34 FZ -0.04
454. 222 FX 0.05 FY -6.79 FZ -2.01
455. ********************************************************************
456. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**260**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**225**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**256**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 174/ 27/ 84 DOF
TOTAL PRIMARY LOAD CASES = 19, TOTAL DEGREES OF FREEDOM = 792
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 67 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.1/ 163658.7 MB

```
457. LOAD LIST 100 TO 109
458. START CONCRETE DESIGN
```

```
459. CODE ACI 1999
460. FC 3567 ALL
461. TRACK 0 ALL
462. MAXMAIN 25 ALL
463. CLB 0.075 MEMB 133 134 201 202
464. CLS 0.075 ALL
465. DESIGN COLUMN 133 134 201 202
```

==

COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 201 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 4932.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

16 - 20 MM 1.396 101 STA 0.700
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 202 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI
---

32 - 12 MM 1.005 100 END 0.700
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

466. END CONCRETE DESIGN
467. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= AUG 6,2020 TIME= 19:11:13 ****

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	04-06-2020	07-08-2020	26-10-2023				Date	04-06-2020	07-08-2020	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

C01

ATTACHMENT 5 STAAD Input & Output File of Foundation Mkd. F4

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    AUG  6, 2020                *
*           Time=    19: 4:43                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F4.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 -1; 14 0 -4 -1.5; 16 0 -4 -0.1; 21 0.75 -4 -0.1; 22 0.75 -4 -0.3
     9. 24 0.5 -4 -0.1; 25 0.5 -4 -0.3; 26 0.25 -4 -0.1; 27 0.25 -4 -0.3; 28 0 -4 -0.3
    10. 29 0.75 -4 -0.5; 31 0.5 -4 -0.5; 32 0.25 -4 -0.5; 33 0 -4 -0.5
    11. 34 0.75 -4 -0.75; 36 0.5 -4 -0.75; 37 0.25 -4 -0.75; 38 0 -4 -0.75
    12. 39 0.75 -4 -1; 41 0.5 -4 -1; 42 0.25 -4 -1; 43 0.75 -4 -1.25; 45 0.5 -4 -1.25
    13. 46 0.25 -4 -1.25; 47 0 -4 -1.25; 68 0.75 -4 -1.5; 70 0.5 -4 -1.5
    14. 71 0.25 -4 -1.5; 72 0.75 -4 -1.75; 74 0.5 -4 -1.75; 75 0.25 -4 -1.75
    15. 76 0 -4 -1.75; 77 0.75 -4 -2; 79 0.5 -4 -2; 80 0.25 -4 -2; 81 0 -4 -2
    16. 82 0.75 -4 -2.2; 84 0.5 -4 -2.2; 85 0.25 -4 -2.2; 86 0 -4 -2.2
    17. 90 -0.25 -4 -0.1; 91 -0.25 -4 -0.3; 92 -0.5 -4 -0.1; 93 -0.5 -4 -0.3
    18. 94 -0.75 -4 -0.1; 95 -0.75 -4 -0.3; 97 -0.25 -4 -0.5; 98 -0.5 -4 -0.5
    19. 99 -0.75 -4 -0.5; 101 -0.25 -4 -0.75; 102 -0.5 -4 -0.75; 103 -0.75 -4 -0.75
    20. 105 -0.25 -4 -1; 106 -0.5 -4 -1; 107 -0.75 -4 -1; 109 -0.25 -4 -1.25
    21. 110 -0.5 -4 -1.25; 111 -0.75 -4 -1.25; 129 -0.25 -4 -1.5; 130 -0.5 -4 -1.5
    22. 131 -0.75 -4 -1.5; 133 -0.25 -4 -1.75; 134 -0.5 -4 -1.75; 135 -0.75 -4 -1.75
    23. 137 -0.25 -4 -2; 138 -0.5 -4 -2; 139 -0.75 -4 -2; 141 -0.25 -4 -2.2
    24. 142 -0.5 -4 -2.2; 143 -0.75 -4 -2.2; 145 0 -1 -2; 146 0 -1 -0.3
    25. 147 0.75 -4 -2.4; 149 0.5 -4 -2.4; 150 0.25 -4 -2.4; 151 0 -4 -2.4
    26. 152 -0.25 -4 -2.4; 153 -0.5 -4 -2.4; 154 -0.75 -4 -2.4; 157 0.75 -4 0.1
    27. 158 0.5 -4 0.1; 159 0.25 -4 0.1; 160 0 -4 0.1; 161 -0.25 -4 0.1
    28. 162 -0.5 -4 0.1; 163 -0.75 -4 0.1
    29. MEMBER INCIDENCES
    30. 133 81 145; 134 28 146
    31. ELEMENT INCIDENCES SHELL
    32. 26 21 24 25 22; 27 24 26 27 25; 28 26 16 28 27; 30 22 25 31 29; 31 25 27 32 31
    33. 32 27 28 33 32; 34 29 31 36 34; 35 31 32 37 36; 36 32 33 38 37; 38 34 36 41 39
    34. 39 36 37 42 41; 40 37 38 13 42; 42 39 41 45 43; 43 41 42 46 45; 44 42 13 47 46
    35. 62 43 45 70 68; 63 45 46 71 70; 64 46 47 14 71; 66 68 70 74 72; 67 70 71 75 74
    36. 68 71 14 76 75; 70 72 74 79 77; 71 74 75 80 79; 72 75 76 81 80; 74 77 79 84 82
    37. 75 79 80 85 84; 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91; 83 92 94 95 93
    38. 85 28 91 97 33; 86 91 93 98 97; 87 93 95 99 98; 89 33 97 101 38
```

```
39. 90 97 98 102 101; 91 98 99 103 102; 93 38 101 105 13; 94 101 102 106 105
40. 95 102 103 107 106; 97 13 105 109 47; 98 105 106 110 109; 99 106 107 111 110
41. 117 47 109 129 14; 118 109 110 130 129; 119 110 111 131 130; 121 14 129 133 76
42. 122 129 130 134 133; 123 130 131 135 134; 125 76 133 137 81
43. 126 133 134 138 137; 127 134 135 139 138; 129 81 137 141 86
44. 130 137 138 142 141; 131 138 139 143 142; 136 82 84 149 147; 137 84 85 150 149
45. 138 85 86 151 150; 139 86 141 152 151; 140 141 142 153 152
46. 141 142 143 154 153; 144 157 158 24 21; 145 158 159 26 24; 146 159 160 16 26
47. 147 160 161 90 16; 148 161 162 92 90; 149 162 163 94 92
48. START GROUP DEFINITION
49. ELEMENT
50. _MOMENT 26 27 30 31 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 71 74 -
51. 75 82 83 86 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 126 127 130 -
52. 131 136 TO 141 144 TO 149
53. _SHEAR 26 30 34 38 TO 40 42 TO 44 62 TO 64 66 70 74 83 87 91 93 TO 95 -
54. 97 TO 99 117 TO 119 123 127 131 136 141 144 149
55. END GROUP DEFINITION
56. ELEMENT PROPERTY
57. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
58. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
59. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 THICKNESS 0.3
60. SUPPORTS
61. 13 14 16 21 22 24 TO 29 31 TO 34 36 TO 39 41 TO 43 45 TO 47 68 70 TO 72 74 -
62. 75 TO 77 79 TO 82 84 TO 86 90 TO 95 97 TO 99 101 TO 103 105 TO 107 -
63. 109 TO 111 129 TO 131 133 TO 135 137 TO 139 141 TO 143 147 149 TO 154 157 -
64. 158 TO 163 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
```

JOINT	INFLUENCE AREA
13	0.062500
14	0.062500
16	0.050000
21	0.025000
22	0.025000
24	0.050000
25	0.050000
26	0.050000
27	0.050000
28	0.050000
29	0.028125
31	0.056250
32	0.056250
33	0.056250
34	0.031250
36	0.062500
37	0.062500
38	0.062500
39	0.031250
41	0.062500
42	0.062500
43	0.031250
45	0.062500
46	0.062500
47	0.062500
68	0.031250

70	0.062500
71	0.062500
72	0.031250
74	0.062500
75	0.062500
76	0.062500
77	0.028125
79	0.056250
80	0.056250
81	0.056250
82	0.025000
84	0.050000
85	0.050000
86	0.050000
90	0.050000
91	0.050000
92	0.050000
93	0.050000
94	0.025000
95	0.025000
97	0.056250
98	0.056250
99	0.028125
101	0.062500
102	0.062500
103	0.031250
105	0.062500
106	0.062500
107	0.031250
109	0.062500
110	0.062500
111	0.031250
129	0.062500
130	0.062500
131	0.031250
133	0.062500
134	0.062500
135	0.031250
137	0.056250
138	0.056250
139	0.028125
141	0.050000
142	0.050000
143	0.025000
147	0.012500
149	0.025000
150	0.025000
151	0.025000
152	0.025000
153	0.025000
154	0.012500
157	0.012500
158	0.025000
159	0.025000
160	0.025000
161	0.025000

```
      162          0.025000
      163          0.012500
   TOTAL           3.750002
 65. DEFINE MATERIAL START
 66. ISOTROPIC STEEL
 67. E 2.09042E+007
 68. POISSON 0.3
 69. DENSITY 7.83341
 70. ALPHA 1.2E-005
 71. DAMP 0.03
 72. TYPE STEEL
 73. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
 74. ISOTROPIC CONCRETE
 75. E 2.21467E+006
 76. POISSON 0.17
 77. DENSITY 2.40262
 78. ALPHA 1E-005
 79. DAMP 0.05
 80. TYPE CONCRETE
 81. STRENGTH FCU 2812.28
 82. END DEFINE MATERIAL
 83. MEMBER PROPERTY AMERICAN
 84. 133 134 PRIS YD 0.55 ZD 0.55
 85. CONSTANTS
 86. MATERIAL CONCRETE ALL
 87. *****ALLOWABLE STRESS DESIGN LOAD COMBS**********************************
 88. LOAD 1 DL+LR
 89. SELFWEIGHT Y -1
 90. ELEMENT LOAD
 91. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
 92. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
 93. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
 94. JOINT LOAD
 95. 145 FX 0.6 FY -3.5 FZ 0.01
 96. 146 FX 0.02 FY -3.32 FZ 0
 97. ************************************************
 98. LOAD 2 DL+0.75*LR+0.75*WL1
 99. JOINT LOAD
100. 145 FX 1.3 FY -1.9 FZ 0
101. 146 FX 0 FY -3.9 FZ 0
102. SELFWEIGHT Y -1
103. ELEMENT LOAD
104. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
105. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
106. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
107. ************************************************************
108. LOAD 3 DL+0.75*LR+0.75*WL2
109. SELFWEIGHT Y -1
110. ELEMENT LOAD
111. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
112. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
113. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
114. JOINT LOAD
115. 145 FX -0.4 FY -4.5 FZ 0
116. 146 FX 0 FY -1.3 FZ 0
117. ************************************************************
```

```
118. LOAD 4 DL+0.75*LR+0.75*WL3
119. SELFWEIGHT Y -1
120. ELEMENT LOAD
121. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
122. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
123. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
124. JOINT LOAD
125. 145 FX 0.5 FY -3.2 FZ 0.05
126. 146 FX 0.02 FY -2.6 FZ 0.04
127. ******************************************************
128. LOAD 5 DL+0.75*LR+0.75*WL4
129. SELFWEIGHT Y -1
130. ELEMENT LOAD
131. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
132. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
133. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
134. JOINT LOAD
135. 145 FX 0.4 FY -3.3 FZ 0.04
136. 146 FX 0.02 FY -2.5 FZ 0.1
137. ******************************************************
138. LOAD 6 0.6*DL+WL1
139. SELFWEIGHT Y -1
140. ELEMENT LOAD
141. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
142. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
143. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -4.4
144. JOINT LOAD
145. 145 FX 1.1 FY 0.4 FZ 0
146. 146 FX 0 FY -2 FZ 0
147. ******************************************************
148. LOAD 7 0.6*DL+WL2
149. SELFWEIGHT Y -1
150. ELEMENT LOAD
151. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
152. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
153. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -4.4
154. JOINT LOAD
155. 146 FX -1.1 FY -3.2 FZ 0
156. 145 FX 0 FY 1.5 FZ 0
157. ******************************************************
158. LOAD 8 0.6*DL+WL3
159. SELFWEIGHT Y -1
160. ELEMENT LOAD
161. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
162. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
163. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -4.4
164. JOINT LOAD
165. 145 FX 0 FY -1.33 FZ 0.06
166. 146 FX 0 FY -0.32 FZ 0.06
167. ******************************************************
168. LOAD 9 0.6*DL+WL4
169. SELFWEIGHT Y -1
170. ELEMENT LOAD
171. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
172. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
173. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -4.4
```

```
174. JOINT LOAD
175. 145 FX 0 FY -1.5 FZ 0
176. 146 FX 0 FY -0.2 FZ 0
177. ********************************************************************************
178. *****ULTIMATE STRENGTH DESIGN LOAD COMBS*************************************
179. LOAD 100 1.4*DL
180. SELFWEIGHT Y -1
181. ELEMENT LOAD
182. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
183. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
184. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5.5
185. JOINT LOAD
186. 145 FX -0.01 FY -3.26 FZ 0
187. 146 FX 0.02 FY -0.58 FZ 0
188. ************************************************
189. LOAD 101 1.4D+1.7(LL)
190. SELFWEIGHT Y -1
191. ELEMENT LOAD
192. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
193. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
194. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5.5
195. JOINT LOAD
196. 145 FX 1.01 FY -5.25 FZ 0.01
197. 146 FX 0.02 FY -5.51 FZ 0
198. ************************************************
199. LOAD 102 1.2D+1.6WL1+1.0L
200. SELFWEIGHT Y -1
201. ELEMENT LOAD
202. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
203. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
204. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
205. JOINT LOAD
206. 145 FX 2.35 FY -1.13 FZ 0.01
207. 146 FX 0.05 FY -6.23 FZ 0
208. ************************************************
209. LOAD 103 1.2D+1.6WL2+1.0L
210. SELFWEIGHT Y -1
211. ELEMENT LOAD
212. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
213. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
214. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
215. JOINT LOAD
216. 145 FX -1.17 FY -6.8 FZ 0.01
217. 146 FX -0.01 FY -0.56 FZ 0.09
218. ************************************************
219. LOAD 104 1.2D+1.6WL3+1.0L
220. SELFWEIGHT Y -1
221. ELEMENT LOAD
222. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
223. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
224. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
225. JOINT LOAD
226. 145 FX 0.65 FY -3.86 FZ 0.1
227. 146 FX 0.02 FY -3.5 FZ -0.09
228. ************************************************8
229. LOAD 105 1.2D+1.6WL4+1.0L
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230. SELFWEIGHT Y -1
231. ELEMENT LOAD
232. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
233. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
234. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
235. JOINT LOAD
236. 145 FX 0.53 FY -4.07 FZ -0.08
237. 146 FX 0.02 FY -3.29 FZ 0
238. ******************************************************************
239. LOAD 106 0.9D+1.6WL1
240. SELFWEIGHT Y -1
241. ELEMENT LOAD
242. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
243. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
244. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
245. JOINT LOAD
246. 145 FX 1.75 FY 0.74 FZ 0
247. 146 FX 0.04 FY -3.21 FZ 0
248. ******************************************************************************
249. LOAD 107 0.9D+1.6WL2
250. SELFWEIGHT Y -1
251. ELEMENT LOAD
252. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
253. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
254. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
255. JOINT LOAD
256. 145 FX -1.77 FY -4.93 FZ 0
257. 146 FX -0.01 FY 2.46 FZ 0.09
258. ******************************************************************************
259. LOAD 108 0.9D+1.6WL3
260. SELFWEIGHT Y -1
261. ELEMENT LOAD
262. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
263. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
264. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
265. JOINT LOAD
266. 145 FX 0.06 FY -1.99 FZ 0.09
267. 146 FX 0.02 FY -0.48 FZ -0.09
268. ******************************************************************************
269. LOAD 109 0.9D+1.6WL4
270. SELFWEIGHT Y -1
271. ELEMENT LOAD
272. 26 TO 28 30 TO 32 34 TO 36 38 TO 40 42 TO 44 62 TO 64 66 TO 68 70 TO 72 74 -
273. 75 TO 76 81 TO 83 85 TO 87 89 TO 91 93 TO 95 97 TO 99 117 TO 119 121 TO 123 -
274. 125 TO 127 129 TO 131 136 TO 141 144 TO 149 PR GY -5
275. JOINT LOAD
276. 145 FX -0.07 FY -2.2 FZ -0.09
277. 146 FX 0.02 FY -0.27 FZ 0
278. ************************************************88
279. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**86**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**66**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**84**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 81/ 12/ 39 DOF
TOTAL PRIMARY LOAD CASES = 19, TOTAL DEGREES OF FREEDOM = 264
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 11 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.5/ 163659.4 MB

280. LOAD LIST 100 TO 109
281. START CONCRETE DESIGN

```
282. CODE ACI 1999
283. TRACK 0 ALL
284. MAXMAIN 25 ALL
285. CLB 0.075 MEMB 133 134
286. CLS 0.075 ALL
287. DESIGN COLUMN 133 134
```

==

COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

288. END CONCRETE DESIGN
289. FINISH

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	04-06-2020	07-08-2020	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C01	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

C01

ATTACHMENT 6 STAAD Input & Output File of Foundation Mkd. F5

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    AUG  6, 2020                *
*           Time=    16:16:36                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F5.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 14 -0.05 -4 -1.5; 16 -0.05 -4 -0.85; 21 0.75 -4 -0.85; 22 0.75 -4 -1.05
     9. 24 0.5 -4 -0.85; 25 0.5 -4 -1.05; 26 0.25 -4 -0.85; 27 0.25 -4 -1.05
    10. 28 -0.05 -4 -1.05; 29 0.75 -4 -1.25; 31 0.5 -4 -1.25; 32 0.25 -4 -1.25
    11. 33 -0.05 -4 -1.25; 34 0.75 -4 -1.5; 36 0.5 -4 -1.5; 37 0.25 -4 -1.5
    12. 72 0.75 -4 -1.75; 74 0.5 -4 -1.75; 75 0.25 -4 -1.75; 76 -0.05 -4 -1.75
    13. 77 0.75 -4 -2; 79 0.5 -4 -2; 80 0.25 -4 -2; 81 -0.05 -4 -2; 82 0.75 -4 -2.2
    14. 84 0.5 -4 -2.2; 85 0.25 -4 -2.2; 86 -0.05 -4 -2.2; 90 -0.3 -4 -0.85
    15. 91 -0.3 -4 -1.05; 92 -0.45 -4 -0.85; 93 -0.45 -4 -1.05; 97 -0.3 -4 -1.25
    16. 98 -0.45 -4 -1.25; 101 -0.3 -4 -1.5; 102 -0.45 -4 -1.5; 133 -0.3 -4 -1.75
    17. 134 -0.45 -4 -1.75; 137 -0.3 -4 -2; 138 -0.45 -4 -2; 141 -0.3 -4 -2.2
    18. 142 -0.45 -4 -2.2; 145 -0.05 -1 -2; 146 -0.05 -1 -1.05; 147 0.75 -4 -2.4
    19. 149 0.5 -4 -2.4; 150 0.25 -4 -2.4; 151 -0.05 -4 -2.4; 152 -0.3 -4 -2.4
    20. 153 -0.45 -4 -2.4; 157 0.75 -4 -0.65; 158 0.5 -4 -0.65; 159 0.25 -4 -0.65
    21. 160 -0.05 -4 -0.65; 161 -0.3 -4 -0.65; 162 -0.45 -4 -0.65
    22. MEMBER INCIDENCES
    23. 133 81 145; 134 28 146
    24. ELEMENT INCIDENCES SHELL
    25. 26 21 24 25 22; 27 24 26 27 25; 28 26 16 28 27; 30 22 25 31 29; 31 25 27 32 31
    26. 32 27 28 33 32; 34 29 31 36 34; 35 31 32 37 36; 36 32 33 14 37; 66 34 36 74 72
    27. 67 36 37 75 74; 68 37 14 76 75; 70 72 74 79 77; 71 74 75 80 79; 72 75 76 81 80
    28. 74 77 79 84 82; 75 79 80 85 84; 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91
    29. 85 28 91 97 33; 86 91 93 98 97; 89 33 97 101 14; 90 97 98 102 101
    30. 121 14 101 133 76; 122 101 102 134 133; 125 76 133 137 81; 126 133 134 138 137
    31. 129 81 137 141 86; 130 137 138 142 141; 136 82 84 149 147; 137 84 85 150 149
    32. 138 85 86 151 150; 139 86 141 152 151; 140 141 142 153 152; 144 157 158 24 21
    33. 145 158 159 26 24; 146 159 160 16 26; 147 160 161 90 16; 148 161 162 92 90
    34. START GROUP DEFINITION
    35. ELEMENT
    36. _MOMENT 26 27 30 31 34 TO 36 66 TO 68 70 71 74 75 82 86 89 90 121 122 126 -
    37. 130 136 TO 140 144 TO 148
    38. _SHEAR 26 30 34 66 70 74 136 144
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39. END GROUP DEFINITION
40. ELEMENT PROPERTY
41. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
42. 122 125 126 129 130 136 TO 140 144 TO 148 THICKNESS 0.3
43. SUPPORTS
44. 14 16 21 22 24 TO 29 31 TO 34 36 37 72 74 TO 77 79 TO 82 84 TO 86 90 TO 93 -
45. 97 98 101 102 133 134 137 138 141 142 147 149 TO 153 157 TO 161 -
46. 162 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
```

JOINT	INFLUENCE AREA
14	0.068750
16	0.055000
21	0.025000
22	0.025000
24	0.050000
25	0.050000
26	0.055000
27	0.055000
28	0.055000
29	0.028125
31	0.056250
32	0.061875
33	0.061875
34	0.031250
36	0.062500
37	0.068750
72	0.031250
74	0.062500
75	0.068750
76	0.068750
77	0.028125
79	0.056250
80	0.061875
81	0.061875
82	0.025000
84	0.050000
85	0.055000
86	0.055000
90	0.040000
91	0.040000
92	0.015000
93	0.015000
97	0.045000
98	0.016875
101	0.050000
102	0.018750
133	0.050000
134	0.018750
137	0.045000
138	0.016875
141	0.040000
142	0.015000
147	0.012500
149	0.025000

```
      150          0.027500
      151          0.027500
      152          0.020000
      153          0.007500
      157          0.012500
      158          0.025000
      159          0.027500
      160          0.027500
      161          0.020000
      162          0.007500
   TOTAL           2.099999
47. DEFINE MATERIAL START
48. ISOTROPIC STEEL
49. E 2.09042E+007
50. POISSON 0.3
51. DENSITY 7.83341
52. ALPHA 1.2E-005
53. DAMP 0.03
54. TYPE STEEL
55. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
56. ISOTROPIC CONCRETE
57. E 2.21467E+006
58. POISSON 0.17
59. DENSITY 2.40262
60. ALPHA 1E-005
61. DAMP 0.05
62. TYPE CONCRETE
63. STRENGTH FCU 2812.28
64. END DEFINE MATERIAL
65. MEMBER PROPERTY AMERICAN
66. 133 134 PRIS YD 0.55 ZD 0.55
67. CONSTANTS
68. MATERIAL CONCRETE ALL
69. *****ALLOWABLE STRESS DESIGN LOAD COMBS**********************************
70. LOAD 1 DL+LR
71. SELFWEIGHT Y -1
72. ELEMENT LOAD
73. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
74. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
75. JOINT LOAD
76. 145 FX -0.2 FY -5.7 FZ 0
77. 146 FX 1.6 FY -7.5 FZ 0
78. ***************************************************
79. LOAD 2 DL+0.75*LR+0.75*WL1
80. JOINT LOAD
81. 145 FX -0.1 FY -4 FZ 1.5
82. 146 FX 1.5 FY -6.6 FZ 0
83. SELFWEIGHT Y -1
84. ELEMENT LOAD
85. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
86. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
87. *******************************************************
88. LOAD 3 DL+0.75*LR+0.75*WL2
89. SELFWEIGHT Y -1
90. ELEMENT LOAD
91. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
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 92. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
 93. *************************************************
 94. JOINT LOAD
 95. 145 FX -0.1 FY -5.6 FZ 0
 96. 146 FX 0.9 FY -5 FZ 0
 97. ************************************************************
 98. LOAD 4 DL+0.75*LR+0.75*WL3
 99. SELFWEIGHT Y -1
100. ELEMENT LOAD
101. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
102. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
103. JOINT LOAD
104. 145 FX -0.2 FY -4.8 FZ 0
105. 146 FX 1.2 FY -5.8 FZ 0
106. ********************************************************
107. LOAD 5 DL+0.75*LR+0.75*WL4
108. SELFWEIGHT Y -1
109. ELEMENT LOAD
110. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
111. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
112. JOINT LOAD
113. 145 FX -0.2 FY -4.8 FZ 0
114. 146 FX 1.2 FY -5.8 FZ 0
115. ********************************************************
116. LOAD 6 0.6*DL+WL1
117. SELFWEIGHT Y -1
118. ELEMENT LOAD
119. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
120. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -4.4
121. ******************************8
122. JOINT LOAD
123. 145 FX 0 FY -0.2 FZ 0
124. 146 FX 0.4 FY -1.4 FZ 0
125. ********************************************************
126. LOAD 7 0.6*DL+WL2
127. SELFWEIGHT Y -1
128. ELEMENT LOAD
129. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
130. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -4.4
131. JOINT LOAD
132. 146 FX -0 FY -2.24 FZ 0
133. 145 FX -0.5 FY 0.7 FZ 0
134. ********************************************************
135. LOAD 8 0.6*DL+WL3
136. SELFWEIGHT Y -1
137. ELEMENT LOAD
138. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
139. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -4.4
140. JOINT LOAD
141. 145 FX 0 FY -1.2 FZ 0
142. 146 FX 0 FY -0.33 FZ 0
143. ********************************************************
144. LOAD 9 0.6*DL+WL4
145. SELFWEIGHT Y -1
146. ELEMENT LOAD
147. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
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148. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -4.4
149. JOINT LOAD
150. 145 FX 0 FY -1.2 FZ 0
151. 146 FX 0 FY -0.33 FZ 0
152. ********************************************************************************
153. *****ULTIMATE STRENGTH DESIGN LOAD COMBS*************************************
154. LOAD 100 1.4*DL
155. SELFWEIGHT Y -1
156. ELEMENT LOAD
157. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
158. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5.5
159. ********************************************************************************
160. JOINT LOAD
161. 145 FX 0 FY -2.8 FZ 0
162. 146 FX 0 FY -0.76 FZ 0
163. ********************************************************************************
164. LOAD 101 1.4D+1.7(LL)
165. SELFWEIGHT Y -1
166. ELEMENT LOAD
167. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
168. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5.5
169. JOINT LOAD
170. 145 FX -0.3 FY -9 FZ 0
171. 146 FX 2.7 FY -13 FZ 0
172. ********************************************************************************
173. LOAD 102 1.2D+1.6WL1+1.0L
174. SELFWEIGHT Y -1
175. ELEMENT LOAD
176. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
177. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
178. JOINT LOAD
179. 145 FX -0.12 FY -4.43 FZ 0
180. 146 FX 2.26 FY -9.32 FZ 0
181. ************************************************
182. LOAD 103 1.2D+1.6WL2+1.0L
183. SELFWEIGHT Y -1
184. ELEMENT LOAD
185. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
186. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
187. ********************************************
188. JOINT LOAD
189. 145 FX -0.18 FY -7.77 FZ 0
190. 146 FX 0.94 FY -5.98 FZ 0
191. ************************************************
192. LOAD 104 1.2D+1.6WL3+1.0L
193. SELFWEIGHT Y -1
194. ELEMENT LOAD
195. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
196. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
197. JOINT LOAD
198. 145 FX -0.15 FY -6.1 FZ 0
199. 146 FX 1.6 FY -7.65 FZ 0
200. ********************************************************************************
201. LOAD 105 1.2D+1.6WL4+1.0L
202. SELFWEIGHT Y -1
203. ELEMENT LOAD
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204. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
205. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
206. JOINT LOAD
207. 145 FX -0.15 FY -6.1 FZ 0
208. 146 FX 1.6 FY -7.65 FZ 0
209. ******************************************************************************
210. LOAD 106 0.9D+1.6WL1
211. SELFWEIGHT Y -1
212. ELEMENT LOAD
213. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
214. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
215. JOINT LOAD
216. 145 FX 0.03 FY -0.13 FZ 0
217. 146 FX 0.66 FY -2.16 FZ 0
218. ******************************************************************************
219. LOAD 107 0.9D+1.6WL2
220. SELFWEIGHT Y -1
221. ELEMENT LOAD
222. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
223. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
224. JOINT LOAD
225. 145 FX -0.03 FY -3.47 FZ 0
226. 146 FX -0.66 FY 1.18 FZ 0
227. ******************************************************************************
228. LOAD 108 0.9D+1.6WL3
229. SELFWEIGHT Y -1
230. ELEMENT LOAD
231. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
232. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
233. JOINT LOAD
234. 145 FX 0 FY -1.8 FZ 0
235. 146 FX 0 FY -0.49 FZ 0
236. ******************************************************************************
237. LOAD 109 0.9D+1.6WL4
238. SELFWEIGHT Y -1
239. ELEMENT LOAD
240. 26 TO 28 30 TO 32 34 TO 36 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 89 90 121 -
241. 122 125 126 129 130 136 TO 140 144 TO 148 PR GY -5
242. JOINT LOAD
243. 145 FX 0 FY -1.8 FZ 0
244. 146 FX 0 FY -0.49 FZ 0
245. ******************************************************************************
246. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**56**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**40**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**54**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 53/ 11/ 36 DOF
TOTAL PRIMARY LOAD CASES = 19, TOTAL DEGREES OF FREEDOM = 174
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 7 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.3/ 163659.6 MB

247. LOAD LIST 100 TO 109
248. START CONCRETE DESIGN

```
249. CODE ACI 1999
250. TRACK 0 ALL
251. MAXMAIN 25 ALL
252. CLB 0.075 MEMB 133 134
253. CLS 0.075 ALL
254. DESIGN COLUMN 133 134
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COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

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COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
28 - 12 MM	1.047	100	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

255. END CONCRETE DESIGN
256. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= AUG 6,2020 TIME= 16:16:50 ****

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* *
* Details about additional assistance from *
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